华沿机器人
HUAYAN ROBOTICS

S-Series Robots

Mechanical Operation Instructions

(For S05/S10/S20/S30/S35/S40/S50 Robots)



Introduction

How to Use the Manual

The Manual is intended for operators of S-Series robots who should have some knowledge of electrical and programming. The manual guides the operator in the use of the robot in the following ways:

- ☑ Safety Considerations: The operators should remember all safety instructions.
- ☑ Mechanical Installation: The operators are instructed on robot installation.
- ➢ Electrical Ports: The open ports of the S-Series robots are introduced to facilitate a secondary development.
- ➢ Software Control: Guide the operator to install and run the software.
- ➢ Safety Configuration: The basic safety settings are introduced.

Technical Support

Guangdong Huayan Robotics Co., Ltd. provides you with long-term technical services. If you have any technical issues or other inquiries during the use of the products, please visit our official website: www.huayan-robotics.com, or consult our engineers directly.

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Content

Introduction ..... **I**
How to Use the Manual I
Technical Support I
Contact Information I
Chapter I Safety Considerations **1**
1.1 Warning Sign 1
1.2 Safety 2
1.3 Responsibility and Specification 6
1.4 Intended Use 8
1.5 Handling of Emergencies 9
1.6 Braking Distance and Braking Time 10
1.7 Handling and Precautions 11
1.8 Other Potential Risks 12
1.9 Maintenance and Repair Instructions 12
Chapter II Product Introduction **13**
2.1 S-Series Robots Introduction 13
2.2 List of Components 13
2.3 Robot Arm 14
2.4 Robot Electrical Control Box 34
Chapter III Mechanical Installation **36**
3.1 Unpacking 36
3.2 Installation Environment Requirements 37
3.3 Installation Space 38
3.4 S05-900 Robot Arm Fixing 38
3.5 End Tool Mounting of S05-900 Robot 40
3.6 S10-1300 Robot Arm Fixing 40
3.7 End Tool Mounting of S10-1300 Robot 41
3.8 S20-1700 Robot Arm Fixing 42
3.9 End Tool Mounting of S20-1700 Robot 43
3.10 S30-1800 Robot Arm Fixing 44
3.11 End Tool Mounting of S30-1800 Robot 45
3.12 S35-1800 Robot Arm Fixing 46
3.13 End Tool Mounting of S35-1800 Robot 48
3.14 S40-1800 Robot Arm Fixing 48

3.16 End Tool Mounting of S40-1800 Robot 50
3.16 S50-2000 Robot Arm Fixing 50
3.17 End Tool Mounting of S50-2000 Robot 52
Chapter IV Quality Assurance 53
4.1 Product Quality Assurance 53
4.2 Disclaimer 53
Chapter V Appendix 55
5.1 Technical Specification 55
5.2 Restriction of Safety-related Functions 62
5.3 Electrical Safety Specification 62

Chapter I Safety Considerations

1.1 Warning Sign

The following different levels of safety signs will be specified in this manual:



DANGER: Failure to follow instructions may result in death or serious injury.



WARNING: Failure to follow instructions may result in injury to personnel or serious damage to equipment.



CAUTION: Failure to follow instructions may result in damage to the equipment.

1.2 Safety

Introduction

The safety principles and specifications that should be observed when operating a robot or robotic system are described in this chapter. The integrator and the user must read this manual in detail, and the contents with warning signs must be mastered and observed strictly. Due to the complexity and hazardous nature of the robot system, the user should be fully aware of the risks involved and strictly observe and implement the specifications and requirements in this manual. The user and the integrator should be fully aware of safety and comply with the safety standard for industrial robots (ISO 10218).

The S-Series robots are equipped with a wide range of built-in safety functions as well as safety I/O, digital and analog control signals to and from the electrical ports for connecting to other machines and additional protection devices.

Note:

1. The use and configuration of safety features and ports must follow the risk assessment procedure for each robot application.
2. If the robot detects a malfunction or violation in the safety system (e.g., a wire is cut in the emergency stop circuit or there is a safety limit violation), the robot should immediately cut off the power and stops
3. The time of stop should be considered as part of the application risk assessment.



Danger:

1. The use of safety configuration parameters that differ from those identified in the risk assessment may result in hazards that cannot reasonably be eliminated or risks that cannot be adequately reduced.
2. Ensure that the tool and clamping jaws are connected correctly to avoid danger in the event of a power failure.
3. The end-effector is not protected by the safety system of Guangdong Huayan Robotics Co., Ltd.. End-effector and/or connecting cable functions are not monitored.



Safety Precautions

Overview

The manual contains safety measures to protect the user and prevent damage to the machine. The user should read all relevant descriptions in the manual and be fully familiar with the safety considerations. We have tried to describe as many situations as possible in this manual. However, there are so many possibilities that it is impossible to document all the situations that cannot be done.

Instruction for Use

The basic information below should be understood and followed when first starting up the robot or robotic system. Other safety-related information is presented in other sections of this manual. However, it is not possible to cover all aspects. In practical application, the specific problems should be analyzed.

Installation and Operation:



1. Be sure to install the robot and all electrical equipment according to the requirements and specifications in this manual.
2. Initial test and inspection of the robot and its protection system are required before the robot is used for the first time and before it is put into production.
3. Before starting the system and equipment for the first time, the equipment and system must be checked for completeness, safe operation and any damage detected. Compliance with valid national or regional safety regulations should be observed during the inspection and all safety features must be tested.
4. The user must check and ensure that all safety parameters and user programs are correct and that all safety functions are functioning correctly. A person qualified to operate the robot is required to check each safety feature. The robot can only be activated after it has passed a thorough and detailed safety test and has reached the safety level.
5. Professionals are required to install and commission the robot in accordance with the installation standards.
6. Once the robot has been installed and built, a full risk assessment should be carried out again and documented.
7. Safety parameters should be set and changed by authorized personnel. Passwords or isolation measures should be adopted to prevent unauthorized personnel from changing or setting the safety parameters. Once the safety factors are modified, the relevant safety functions should be analyzed.

8. The robot can be stopped by pressing the emergency stop button in the event of an accident or abnormal operation.

9. Brakes are mounted inside the robot joint module to maintain its attitude in the event of a power failure. Do not turn on and off the power supply system frequently. It is recommended that the interval between turning on and off should be more than 10s.

10. The robot is equipped with collision detection function. The robot automatically stops when the external force applied to the robot exceeds the normal force range set by the safety settings of the user to prevent injury to the robot or operator from collision. This function is specifically designed for S-Series robots for safe human-robot co-operation. But it requires that the robot system should be within normal operating range and that an S-Series electrical control box be used. If the controller is developed by the user, the robot will not have the above functions. Any consequences of the resulting danger should be borne by the user.

Operational and Heat Generation Aspects

1. The robot arm and the electrical control box will generate heat during the operation. Do not operate or touch the robot while it is operating or when it is temporarily stopped.



2. Turn off the power and wait one hour for the robot arm to cool down.

3. Do not put your fingers in the area where the electrical control box is generating heat.

Usage

1. Ensure that the robot arm and tools are properly and securely in place.



2. Ensure that the robot arm has enough space to move freely.

3. Do not use the robot if it is damaged. Please contact the technical support engineers of Guangdong Huayan Robotics Co., Ltd.

4. Do not connect safety devices to normal I/O ports; Only the safety-type I/O ports can be used.

5. Ensure that the correct installation settings are made (e.g., installation angle of the robot arm, weight in TCP, TCP tool coordinates, safety configuration). Save and load the installation file into the program.

6. Tools and obstacles should not have sharp corners or twisted points. Ensure that the heads and the faces of all persons are out of reach of the robot.

7. Pay attention to the movement of the robot when using the teach pendent. Any impact will release a large amount of kinetic energy that is much higher than that in the case of high speed and high payload.

8. Connecting different pieces of machinery may aggravate the hazard or create a new one. A full

risk assessment should be always carried out for the entire installation. When different safety and emergency stop performance levels are required, the highest performance level should be selected. Keep reading and understanding the manual for all equipment used in the installation.

9. Do not modify the robot. Changes to the robot could potentially create hazards that the integrator could not have predicted. Authorized robot restructuring is subject to the latest version of all relevant service manuals. If the robot is altered or modified in any way, Guangdong Huayan Robotics Co., Ltd. declines all responsibilities.

Application Environment

1. When the robot is connected to or works with machinery capable of causing damage to the robot, it is strongly recommended that all functions of the robot as well as the robot programs be checked separately. It is recommended that a temporary space other than the mechanical workspace be used to test the robot programs.

2. Guangdong Huayan Robotics Co., Ltd. is not responsible for any damage to the robot or injury to personnel caused by script errors or improper operation of the robot.

3. Do not expose the robot to permanent magnetic fields at all times. Strong magnetic field may damage the robot.



Personnel Safety

When running a robotic system, the safety of the operator must first be ensured. The following is a list of general considerations. Please take appropriate measures to ensure the safety of workers.

Each operator using the robotic system should receive training through a training course hosted by Guangdong Huayan Robotics Co., Ltd.



The user should be fully qualified to operate the robot by mastering the safe and standardized operating procedures. For more information about the training, please contact us on our website:http://www.hansrobot.net

1. Each operator using the robot system is requested to refrain from wearing loose-fitting clothing and jewellery. Ensure that the long hair is bound at the back of your head when operating the robot.

2. Even if the robot appears to have stopped while the equipment is in running, it may be in a state of impending action because it is waiting for a start signal. Even in such a state, the robot should be considered to be in movement.



3. Lines should be drawn on the floor to mark the robot's range of movement so that the operator understands that the robot contains the range of movement of the holding tool (manipulator, tool).

4. Ensure that safety measures (e.g., guardrails, ropes or protective screens) are in place near the operating area of robot to protect the operator and the surrounding population. Locking devices should be installed as necessary so that the robot power supply cannot be accessed by anyone other than the operator in charge of the operation.

5. When using the operation panel and the teach pendant, be sure to work with gloves removed, as there may be operating errors when wearing gloves.

6. Force the joints to move by pushing or pulling the robot arm with force (at least 500N) in emergency and abnormal situations, such as when a person is pinned or surrounded by the robot. Moving the robot arm manually without electric power is limited to emergency and may damage the joint.

1.3 Responsibility and Specification

S-Series robots can be combined with other equipment to form complete machines, which are not complete in themselves. Thus, the information in this manual does not contain the information on how to fully design, install and operate a complete robot, nor does it contain all the possibilities of influencing the safety of the peripheral equipment of the complete system. The safety of installing a complete robot depends on how the robot is integrated. The integrator should conduct a risk assessment of the design and installation of the complete system in compliance with the laws and regulations of the host country as well as safety specifications and standards. Risk assessment is one of the most important tasks that must be done to integrate business. The integrator could implement the risk assessment process with reference to the standard below. Risk assessment is one of the most important tasks that must be done to integrate business. The integrator could implement the risk assessment process with reference to the standard below.

- ISO 12100:2010 Safety of Machinery — General Principles for Design — Risk Assessment and Risk Reduction.
- ISO 10218-2:2011 Robots and Robotic Devices — Safety Requirements for Industrial Robots - Part 2: Robot Systems and Integration.
- RIA TR R15.306-2014 Technical Report on Industrial Robots and Robotic Systems - Safety Requirements, Task-based Risk Assessment Methodology.
- ANSI B11.0-2010 Machinery Safety - General Requirements and Risk Assessment.



The robot integrator is required to fulfil, but not limited to, the following responsibilities:

- Conduct a full risk assessment for the complete robotic system and confirm that the entire system is designed and installed accurately;
- Provide training to users and staff;
- Create a complete and systematic Code of Conduct with a clear description of the processes to be used, and establish the appropriate safety measures;
- Eliminate the hazard or minimize all hazards to an acceptable level using appropriate methods at the time of final installation;
- Communicate the residual risk to the end user;
- Label the robot with the logo and contact information of the integrator, and archive the relevant technical documentation.
- For access to applicable standards and legal guidelines, please visit the website below: www.hansrobot.net
- All safety-related information contained in this manual is not to be regarded as a warranty by Guangdong Huayan Robotics Co., Ltd. Even if all safety instructions are followed, injury to persons or damage to equipment caused by the operator may still occur.
- Guangdong Huayan Robotics Co., Ltd. is committed to continuously improving the reliability and performance of its products and therefore reserves the right to upgrade its products without prior notice. Guangdong Huayan Robotics Co., Ltd. seeks to ensure the accuracy and reliability of the contents of this manual, but is not responsible for any errors or omissions in the information contained herein.

Hazard Identification



For the risk assessment, all potential contact between the operator and the robot during normal use as well as foreseeable misuse should be considered. The neck, face and head of an operator should not be exposed to avoid touching. The use of a robot without the use of peripheral safety guards requires a risk assessment to be carried out first to determine whether the hazard in question poses an

unacceptable risk, for example:

- The use of sharp end-effector or tool connector may be hazardous;
- Handling of toxic or other hazardous substances may be hazardous;
- The fingers of an operator are in danger of getting caught in the robot base or joint;
- Danger of collision by robot;
- Danger of poor fixing of the robot or tools attached to the end-effector;
- Hazards caused by impacts between the robot payload and solid surfaces.

The integrator must measure such hazards and their associated risk levels through a risk assessment and identify and implement appropriate measures to reduce the risk to an acceptable level. Please note that there may be other significant hazards associated with specific robotic devices.

The risks associated with the operation of the S-Series robots are minimized to as low a level as reasonably practicable by combining the inherent safety design measures applied to the robots with safety specifications or risk assessments carried out by integrators and end users. This document allows the user to communicate any residual risk that exists for the robot prior to installation to the integrator and end user. If the risk assessment conducted by the integrator determines that there are hazards in its particular application that may pose an unacceptable risk to the user, the integrator must take appropriate risk reduction measures to eliminate or minimize those hazards until the risk is reduced to an acceptable level. It is not safe to use until appropriate risk reduction measures have been taken (if required).

In the case of a non-cooperative installation of a robot (e.g., when hazardous tools are used), the risk assessment may infer that the integrator needs to connect additional safety equipment (e.g., a safe start-up device) at the time of its programming to ensure the safety of personnel and equipment.

1.4 Intended Use

The application statement below is made when the robot is used only as a general industrial device, such as used to operate or hold tools or equipment, and to process or transfer parts or products. The robot is only permitted to be used under specified environmental conditions. For specific information on the operating environment and operating conditions, please refer to the Appendix.

The robot has special safety level characteristics, but only if they have been risk-assessed to be non-hazardous, i.e., without the use of any safety guards and field sensing devices, expected or accidental

contact between personnel and the robot or its end-effector or components does not constitute an unacceptable risk, nor does the expected or accidental contact with other objects in the work area (tools equipment, surfaces) constitute an unacceptable risk.

Robot controller and robot are limited to general industrial equipment use only and may not be used in applications contrary to their intended use, and prohibited uses include, but are not limited to, the following:



- For the hazardous environments such as flammables and explosives;
- For the devices moving or carrying people or other animals;
- For the devices such as medical equipment involving human life;
- For the devices with significant social and public impact;
- For the vibration environments such as vehicles and ships;
- For the use of climbing tools.

1.5 Handling of Emergencies



Emergency Stop Device:

Pressing the emergency stop button will stop all movements of the robot. Emergency shutdown cannot be used as a risk reduction measure, but can be used as a secondary protection device. If more than one emergency stop button has to be connected, this must be included in the risk assessment of the robot application. The emergency stop button meets the requirements of IEC 60947-5-5.

The robot has an emergency stop buttons in the electrical control box and on the teach pendant, which must be pressed in case of danger or emergency. The electrical control box is equipped with an external emergency stop button port, which can be used by the integrator or the user according to the actual situation.

The tools or equipment connected to the end-effector that pose a potential threat must be integrated into the emergency stop circuit of the system. Failure to comply with this warning may result in death, serious personal injury or substantial property damage.

Recovery from Emergency:

All emergency stop devices in push-button form have a "Locking" function. This "Lock" must be opened to end the emergency stop of the device.

Rotate the emergency stop button to open the "Lock".



Recovering from an emergency stop is a simple but very important step, which should only be performed after ensuring that the robot system is completely free of hazards.



Forced Emergency Movement of the Joint:

In rare cases, it may be necessary to move one or more robot joints in an emergency situation where the power supply to the robot fails or the power supply is not used, which can be forced to move the robot joints by the following methods:

Forced reverse drive: Push or pull the robot arm with force (at least 500N) to force the joints to move.

Forced manual movement of the robot arm is limited to emergency and has the potential to damage the joint.



Robot Arm Over-energy Safety Protection:

The robot arm has the over-energy safety protection function. When an operator or other object accidentally touches the robot arm while it is powered on and stationary, and the collision force exceeds the safety valve value, the robot arm will move passively in the direction of the collision force. This function ensures that in the event of a collision between an operator or other object and the robot arm, injuries to personnel, other objects, and the robot arm are reduced.

This function reduces crash injury and is subject to risk assessment for other applications. Handling of safety emergency.

1.6 Braking Distance and Braking Time

Braking distance and braking time for category 1 stop

The table below shows the braking distance and braking time measurements when the robot triggers a category 1 stop in the configurations below:

Extension: 100% (the robot fully extends horizontally at a maximum arm length).

Velocity: 100% (the general velocity of robot is set to 100% to execute movements at a joint speed of 150 °/s).

Payload: The maximum payload handled by the robot connected to TCP.

Position	Braking Distance (rad)	Braking Time (ms)
1 axis (Base)	0.281	288
2-axis (Shoulder)	0.236	280
3-axis (Elbow)	0.460	327

1.7 Handling and Precautions

When the robot carrying out lifting operation, the moving parts should be positioned with appropriate measures so that they do not create unintended movements that could cause harm during lifting and transport.

Packaging for transport should be carried out in accordance with packaging standards and with the required markings on the outside of the box. When transporting the robot, please ensure that it is stable and that it remains fixed in the proper position. The electrical control box should be lifted using the handle. When moving the robot to the installation position from the packing materials of robot, the robot should be held until all the bolts of the robot's base are fully tightened.



Once fixed, the robot is powered on and the robot attitude is adjusted to a suitable position using the robot Free Drive function. The original packaging should be maintained properly upon completing the transportation. Store the packaging materials in a dry place if repackaging is required and move the robot in the future.

Ensure that you are not overloading your back or other body parts when lifting the equipment.

All regional and national guidelines should be observed. Guangdong Huayan Robotics Co., Ltd. is not responsible for damages incurred during the transport of the equipment.



Ensure that the installation instructions in the manual are strictly observed when installing the robot.

1.8 Other Potential Risks

Please note that there may be other significant hazards associated with specific robotic devices:

- Fingers are trapped between the robot legs and the machine base.
- The fingers are trapped between the large-arm and small-arm.
- Sharp edges and points on tools or tool connectors pierce skin.
- Sharp edges and points on obstacles near the trajectory of robot pierce the skin.
- Injured by robot collision.
- Sprains or fractures due to impact between the robot payload and a solid surface.
- Consequences caused by loosening bolts used to hold robot arms or tools.
- Objects fall off the tool, e.g. due to poor clamping or power failure.
- Operating errors due to different emergency stop buttons on different machines.

1.9 Maintenance and Repair Instructions

It is important that all safety instructions in this manual are strictly observed for maintenance and repair work.



Danger:

- Remove the main input cable from the bottom of the electrical control box to ensure that it is completely disconnected. Disconnect any other energy sources connected to the robot arm or electrical control box. Take necessary precautions to prevent others from reconnecting system energy during maintenance.
- Check the grounding connection before turning on the system again.
- Observe ESD regulations when disassembling the robot arm or electrical control box.
- Avoid splitting the power supply system inside the electrical control box. The power supply system retains high voltage (up to 600 V) for several hours after the electrical control box is turned off.
- Avoid the ingress of water or dust into the robot arm or electrical control box.
- Upon the maintenance and repair, it must be checked to ensure the level of safety required for the service. When checking, the effective national or local safety laws and regulations must be observed. All safety functions should also be tested to check if they are all functioning properly.

Chapter II Product Introduction

2.1 S-Series Robots Introduction

S series robot is a new type of six-axis robot, interacting and coordinating with humans in the work area, its overall design has a lightweight, modular, compact structure, beautiful appearance and other characteristics, can be flexibly adapted to a variety of complex and changing industrial production environment.

S series robots are programmed in drag-and-drop mode, and the program is also drag-and-drop coded, which detaches the technician from the complicated and cumbersome programming demonstration, and the requirement for the user is almost zero threshold. In terms of safety, the robot can be stopped at a faster speed to minimize more serious damage by limiting and detecting torque and power when in collision and contact with people or other objects.

The robot has a smooth and rounded appearance, which does not have sharp corners, thus effectively protecting operator or robot.

2.2 List of Components

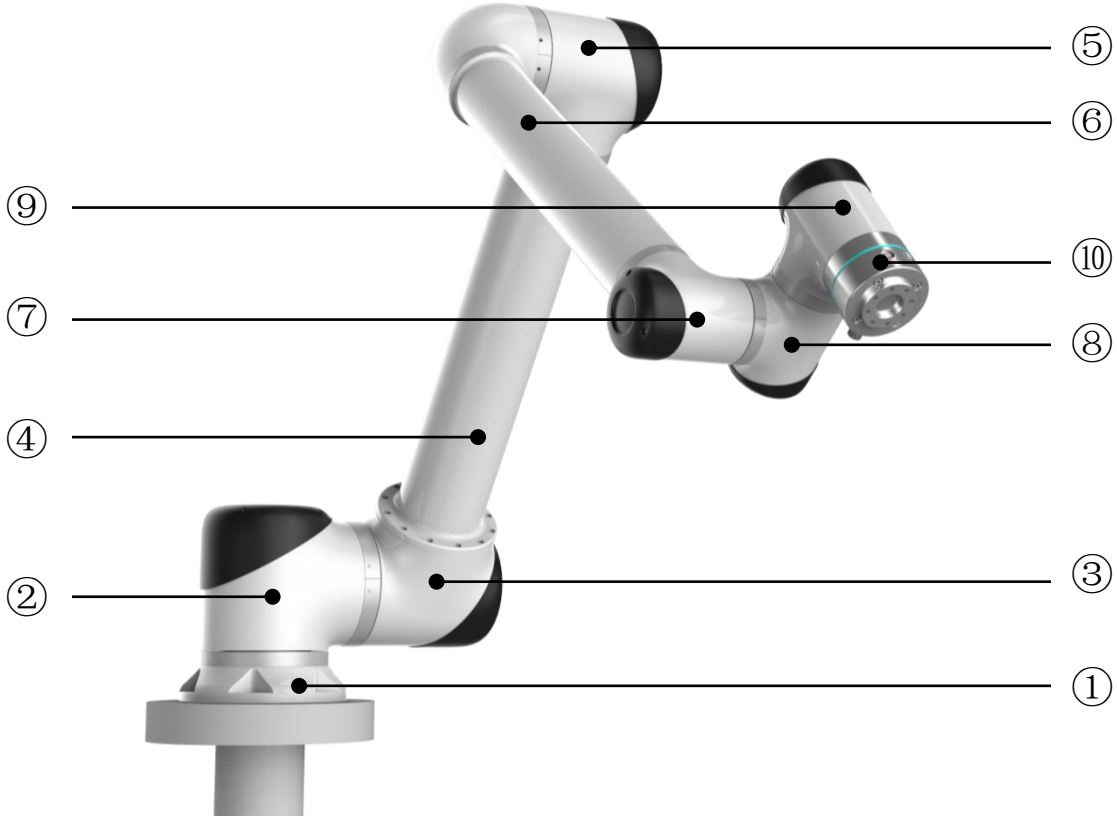
List of Components for the Complete Robot Arm System

Robot	Underwater Robot Arm	×1
	End Connecting Wire	×1
Electrical control box	Electrical Control Box	×1
	Power Cable Wire	×1
	Connecting Cable Wire	×1
Teach pendent	Teach Pendent	×1
User manual	Saved in USB	×1

The above list will be different according to the robot models and customer requirements, such as the electrical control box can also be the one with smaller volume or the lighter mini electrical control box. The above configuration list is for reference only.

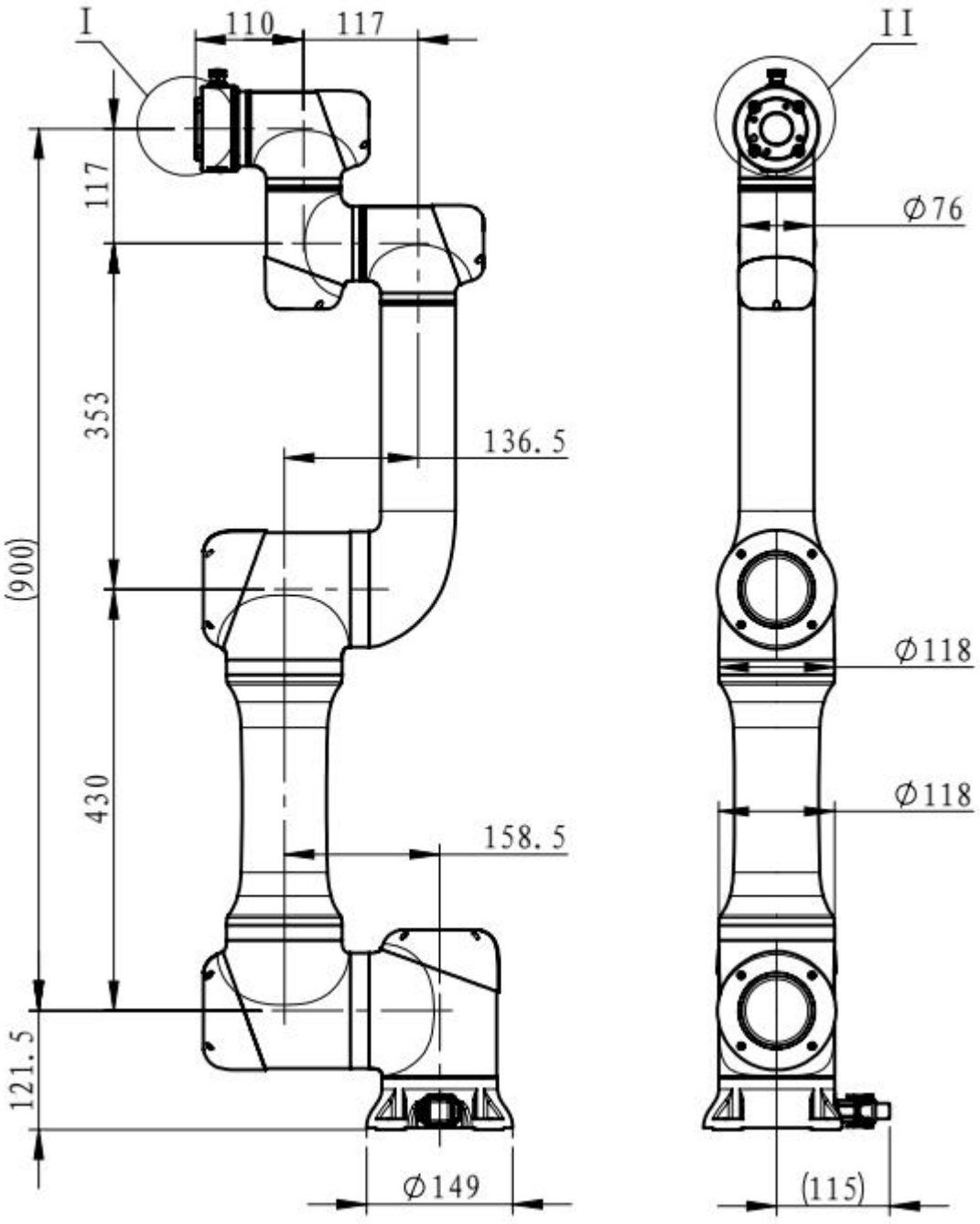
2.3 Robot Arm

2.3.1 Names of the Robot Parts

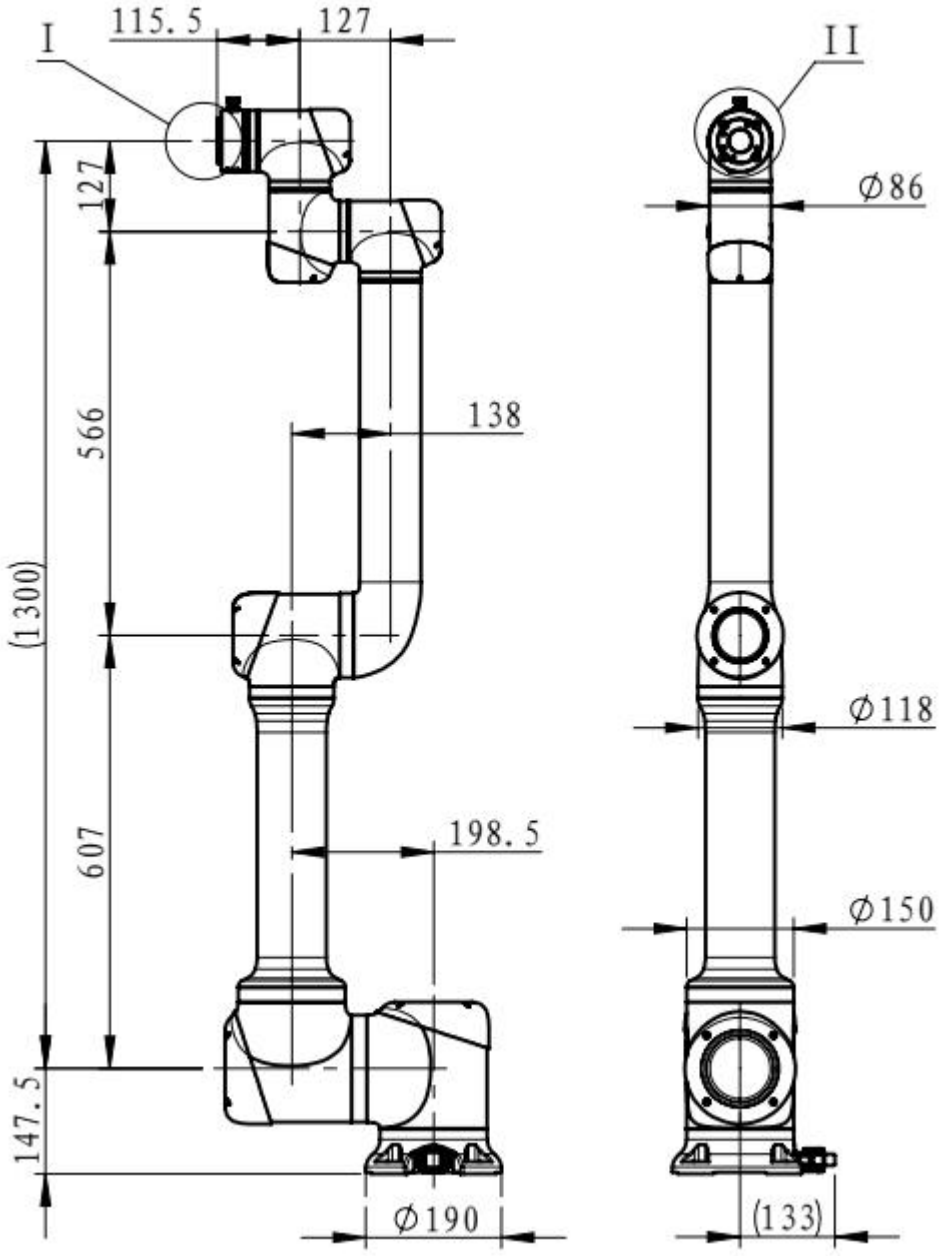


① Machine Base ② J1 Shaft ③ J2 Shaft ④ Lower Arm ⑤ J3 Shaft ⑥ Upper Arm ⑦ J4 Shaft ⑧ J5 Shaft ⑨ J6 Shaft ⑩ End Flange

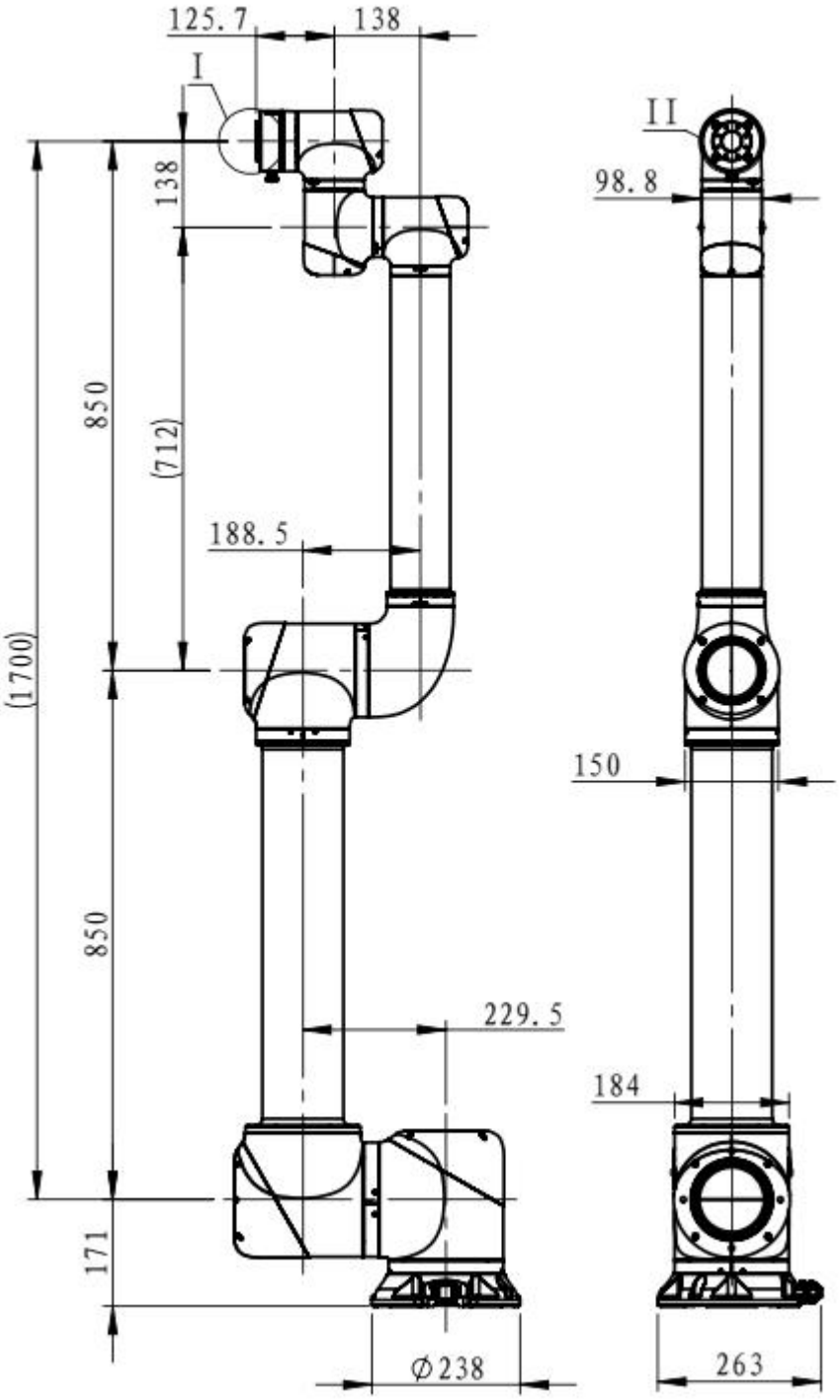
2.3.2 Overall Dimension



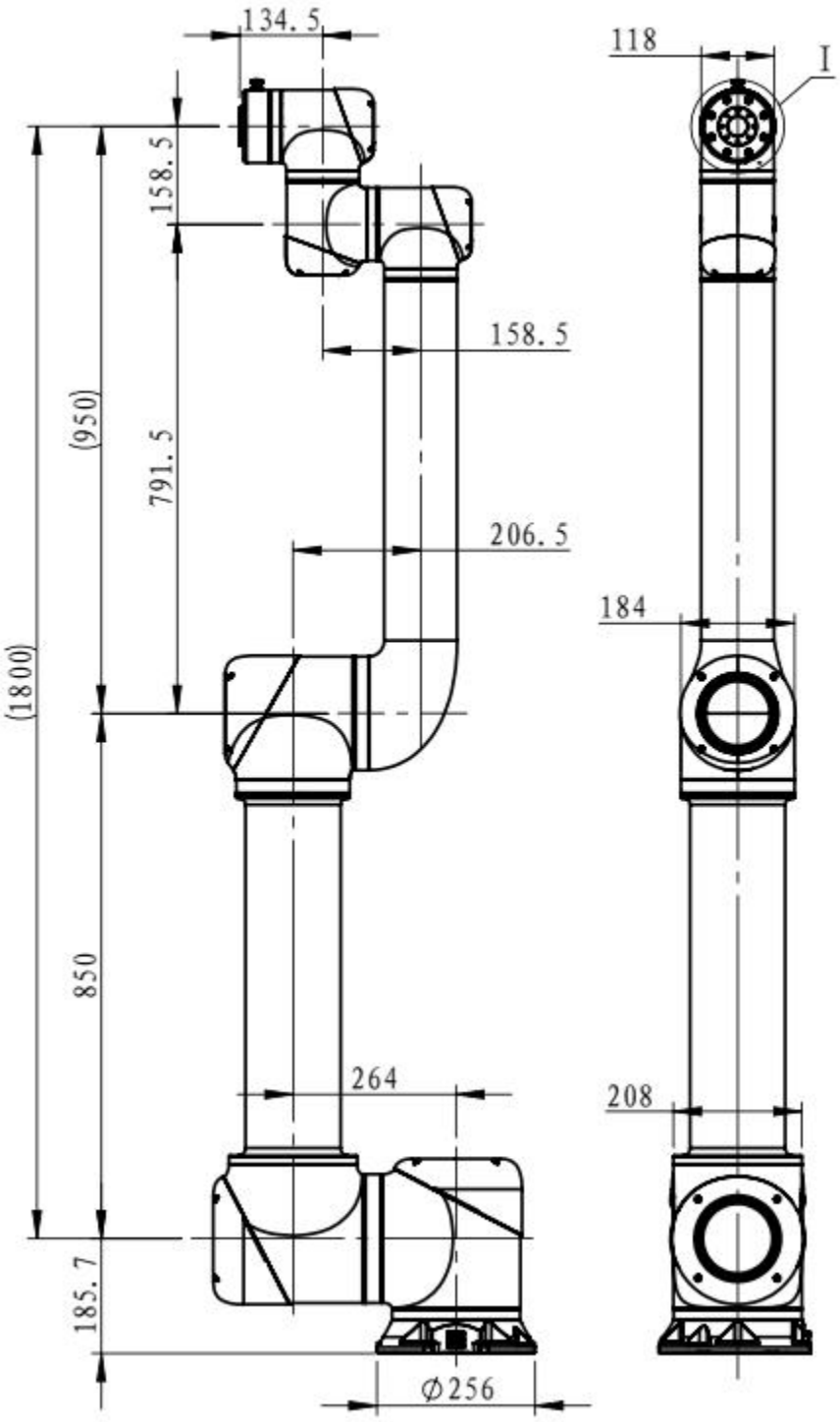
Overall Dimension of S05-900 Manipulator



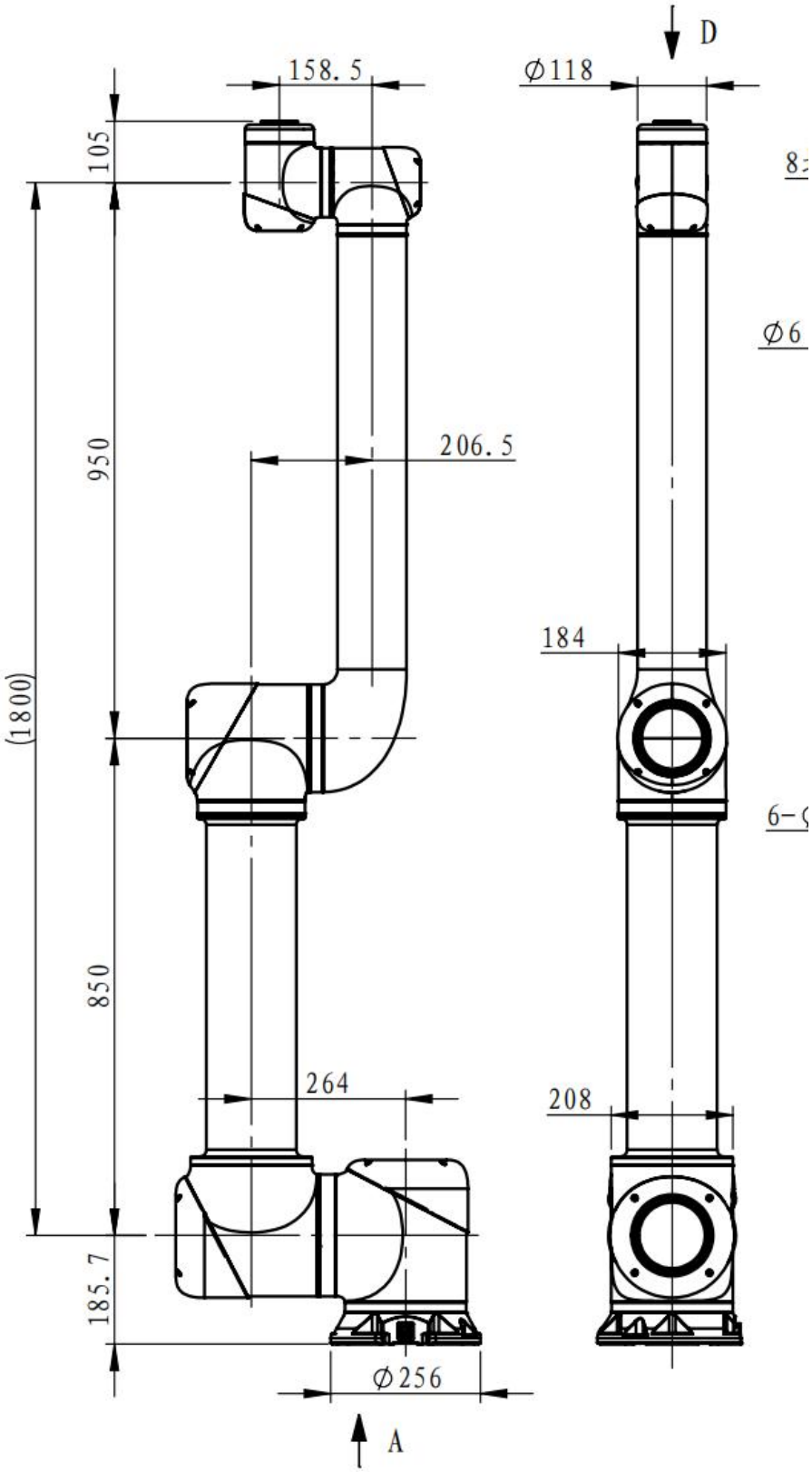
Overall Dimension of S10-1300 Manipulator



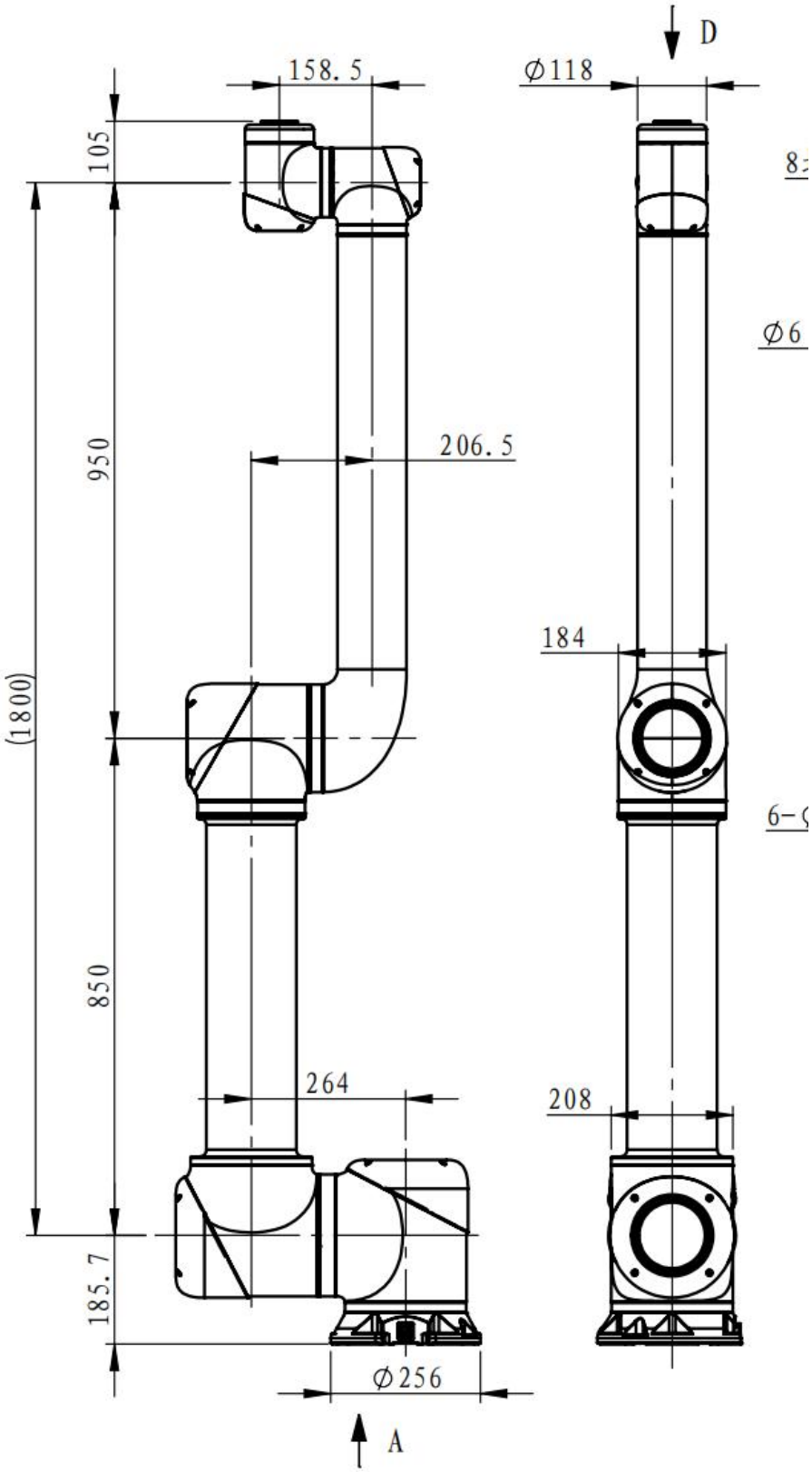
Overall Dimension of S20-1700 Manipulator



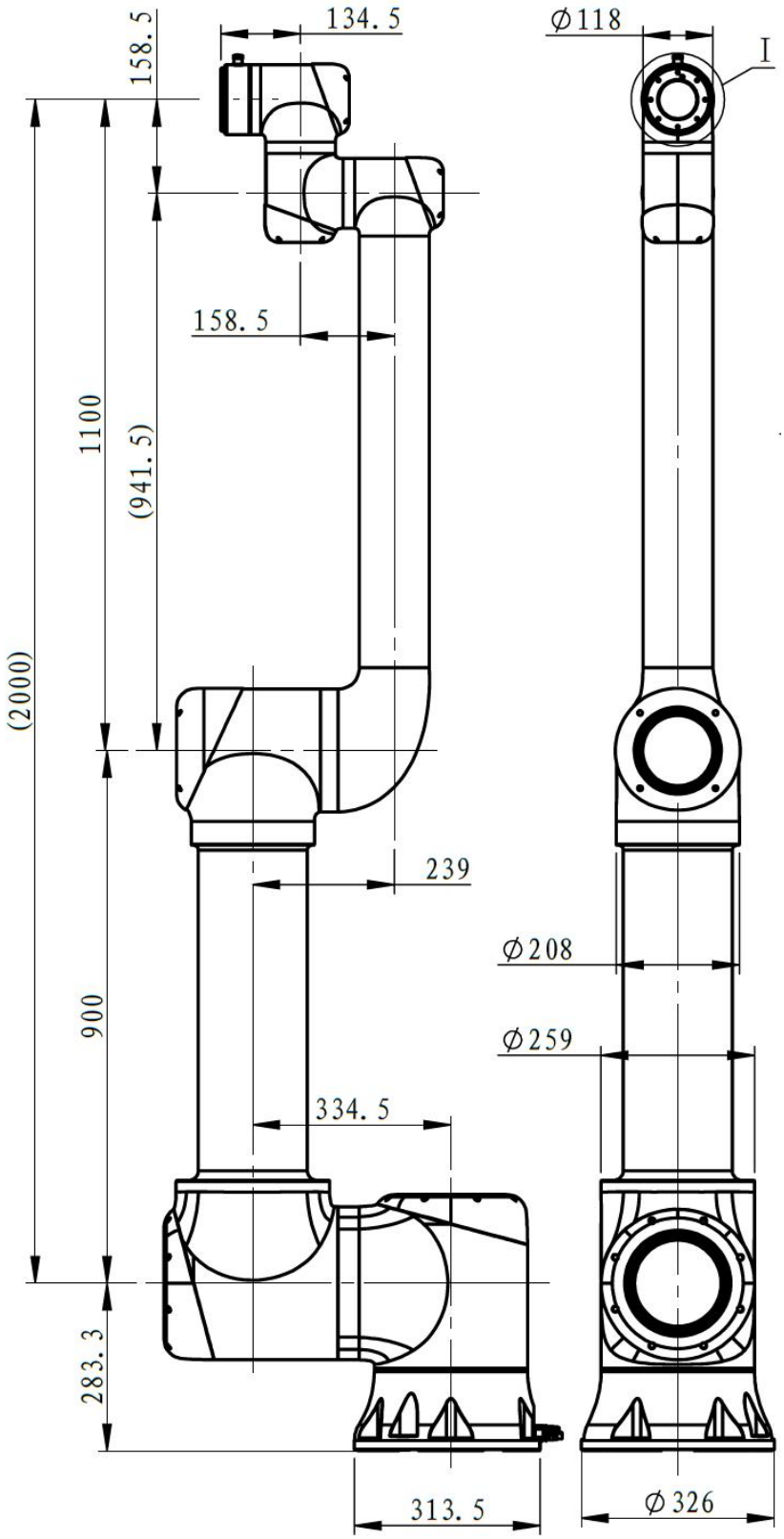
Overall Dimension of S30-1800 Manipulator



Overall Dimension of S35-1800 Manipulator



Overall Dimension of S40-1800 Manipulator



Overall Dimension of S50-2000 Manipulator

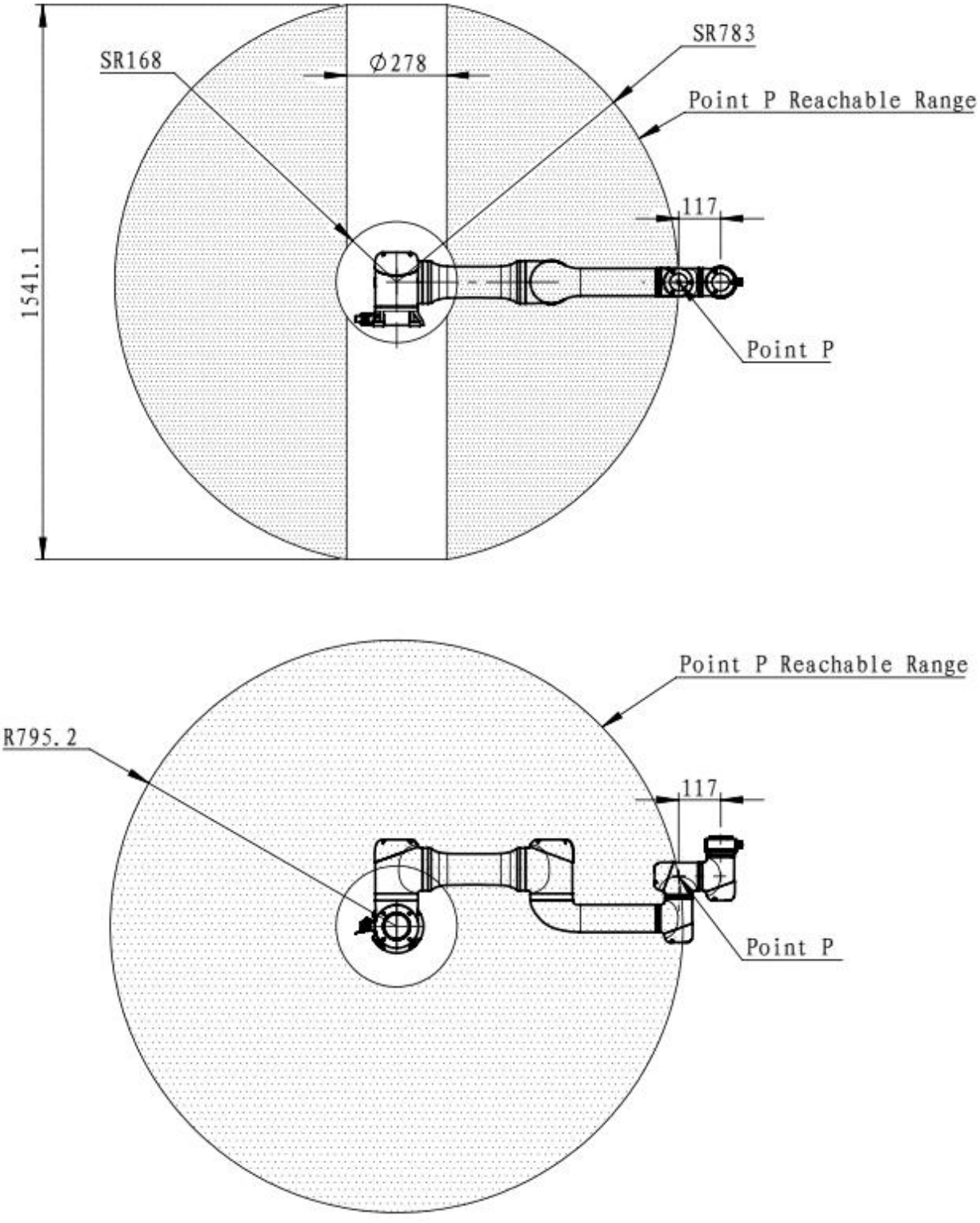
2.3.3 Robot Working Areas

The robot workspace is the area within a specified range around the machine base joints (as shown in the table below). When selecting a robot installation location, be sure to consider the cylinder space directly above and below the robot.

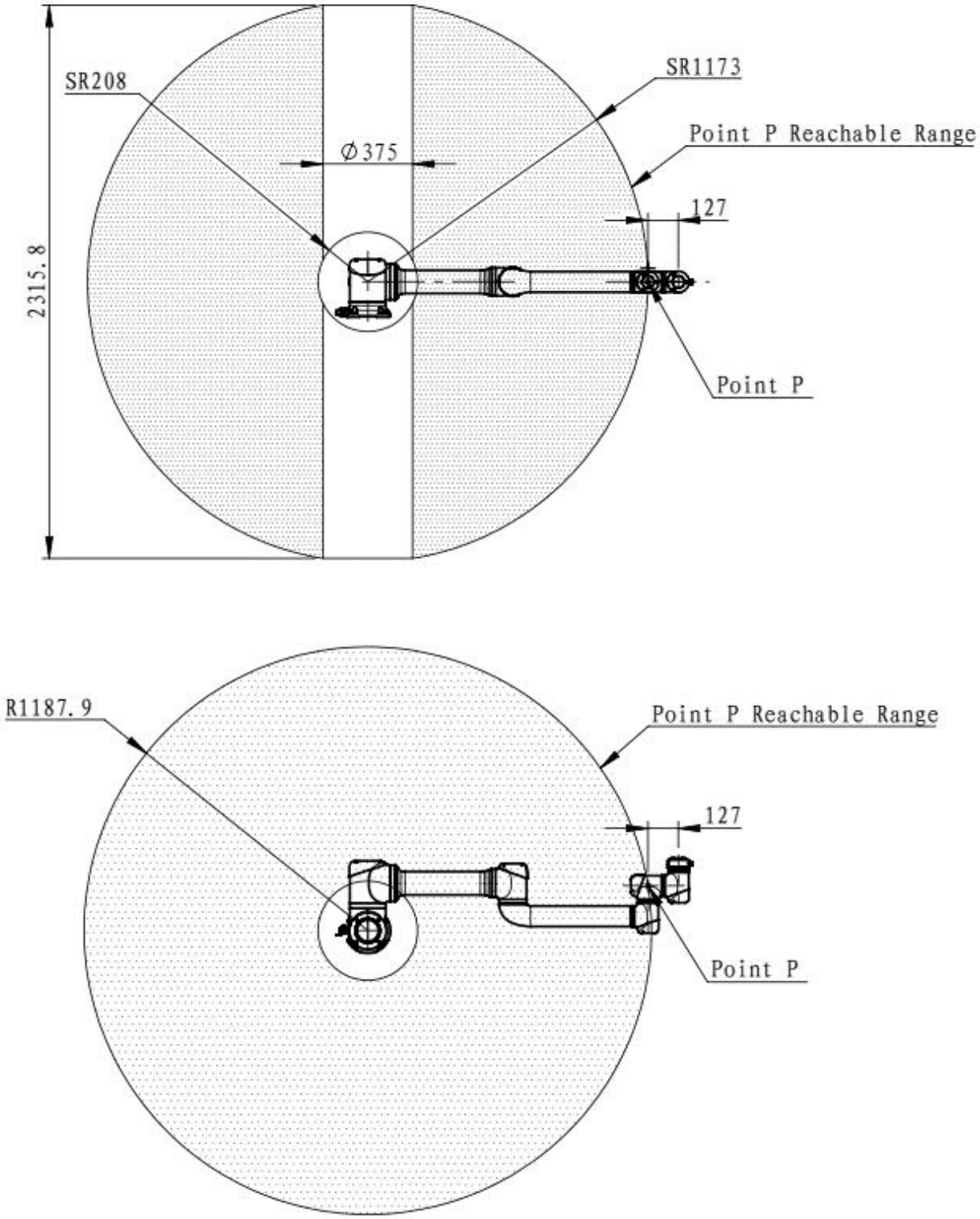
Whenever possible, avoid moving the tool into cylindrical space, as this can cause the joints to move too fast while the tool is moving slowly, resulting in low working efficiency of the robot and difficult risk assessment.

Operating Radius of Robot

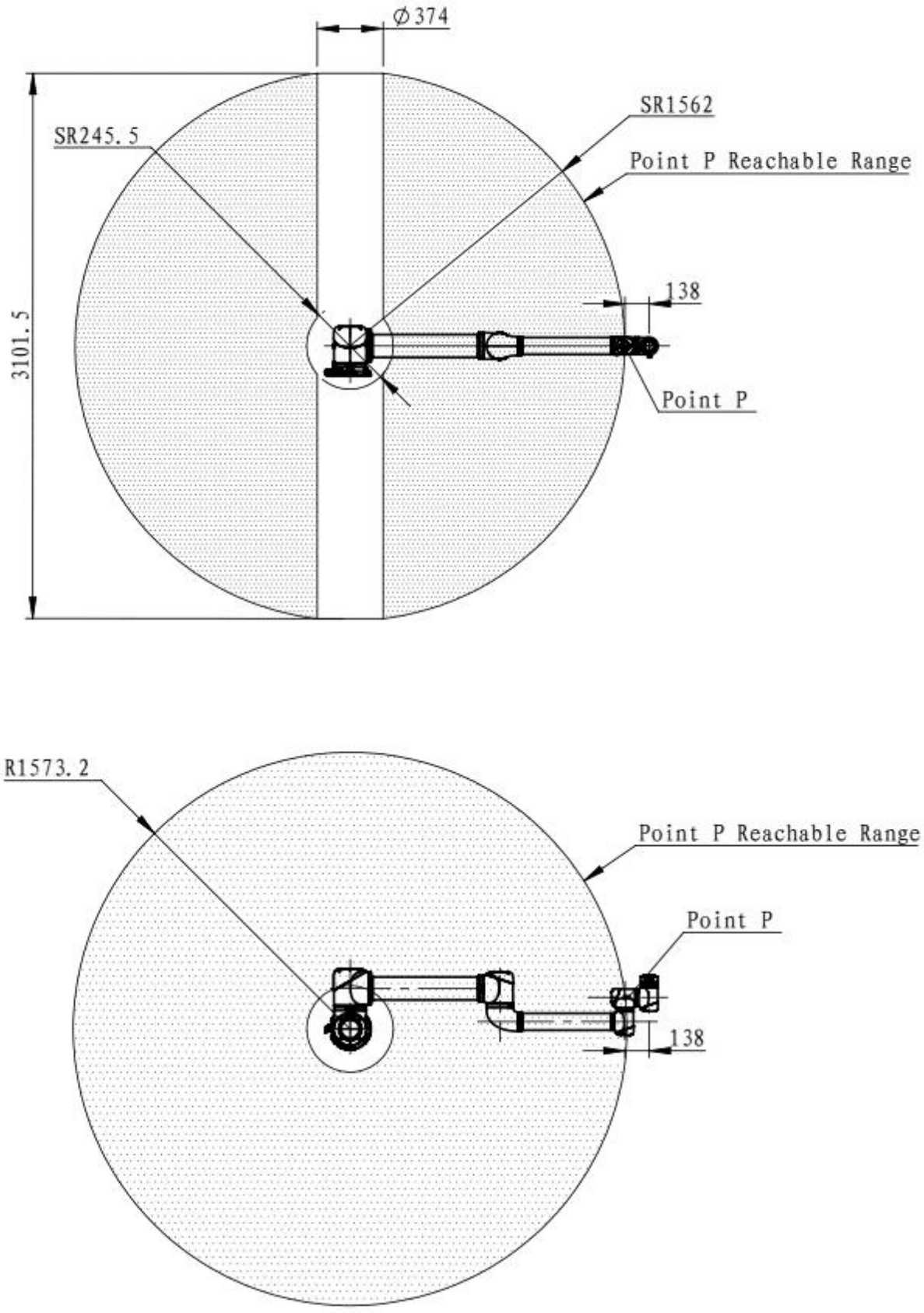
Model	Operating Radius
S05	900 mm
S10	1300 mm
S20	1700 mm
S30	1800 mm
S35	1800mm
S40	1800mm
S50	2000mm



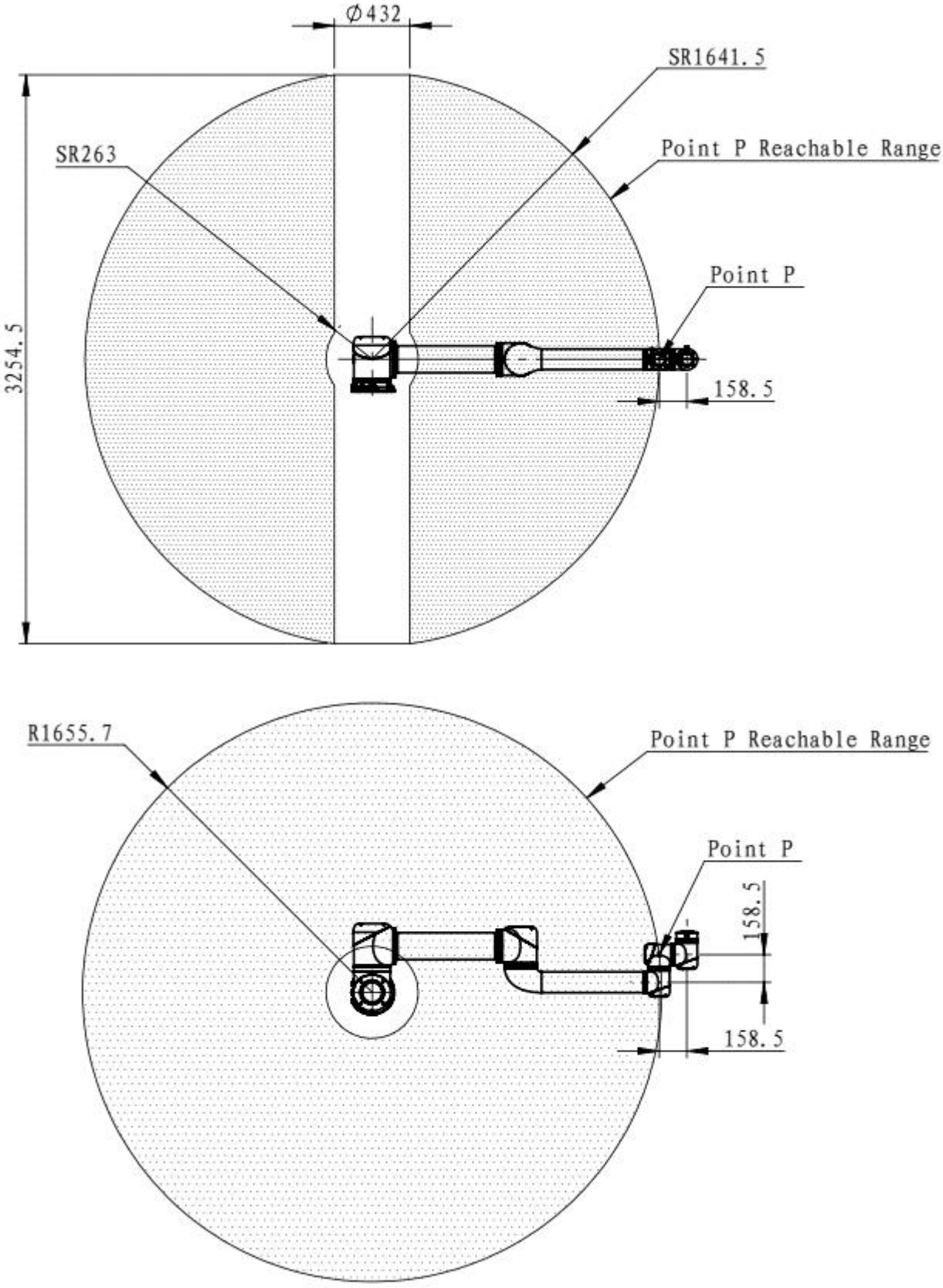
Movement Range of S05-900 Robot



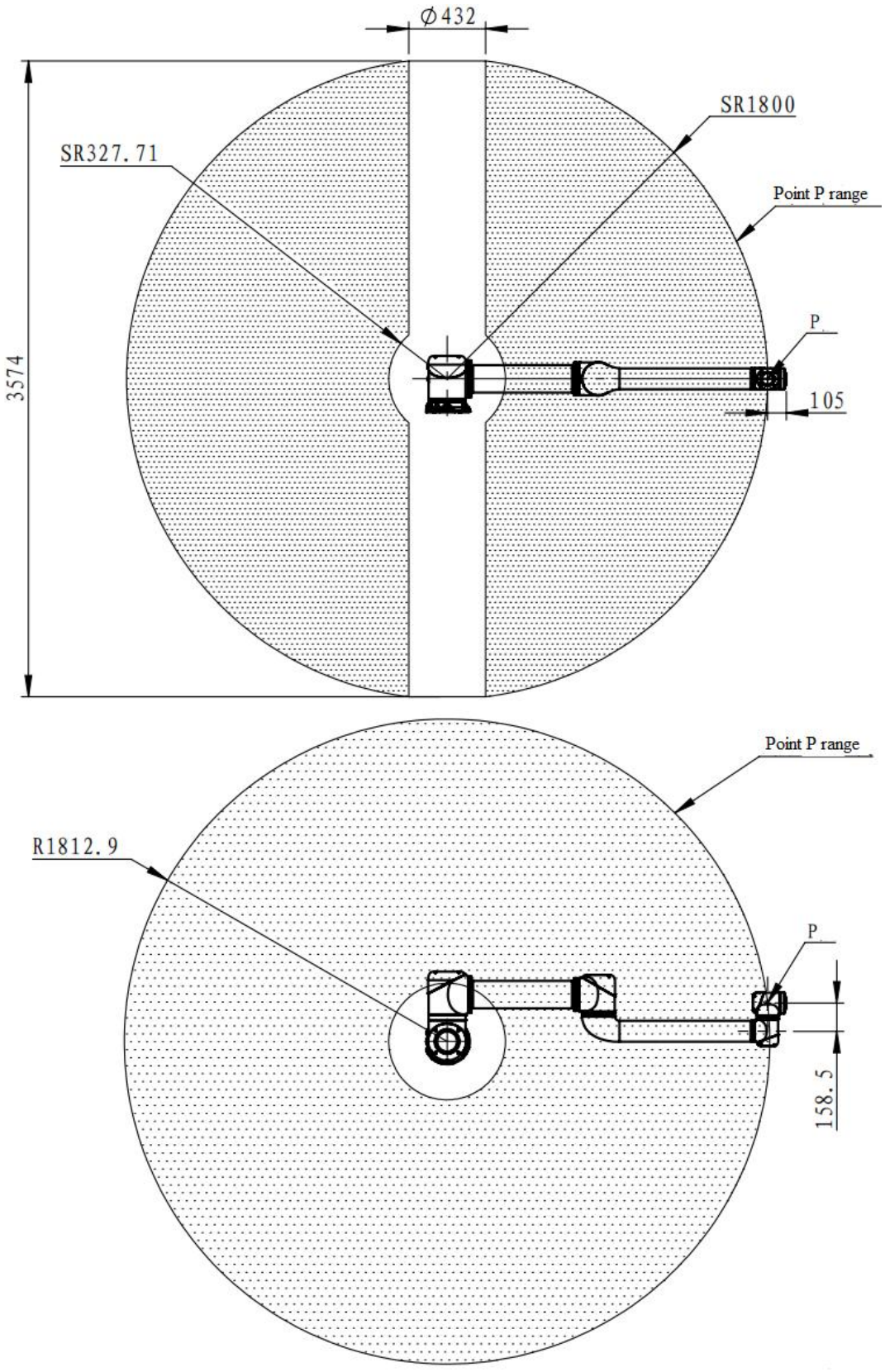
Movement Range of S10-1300 Robot



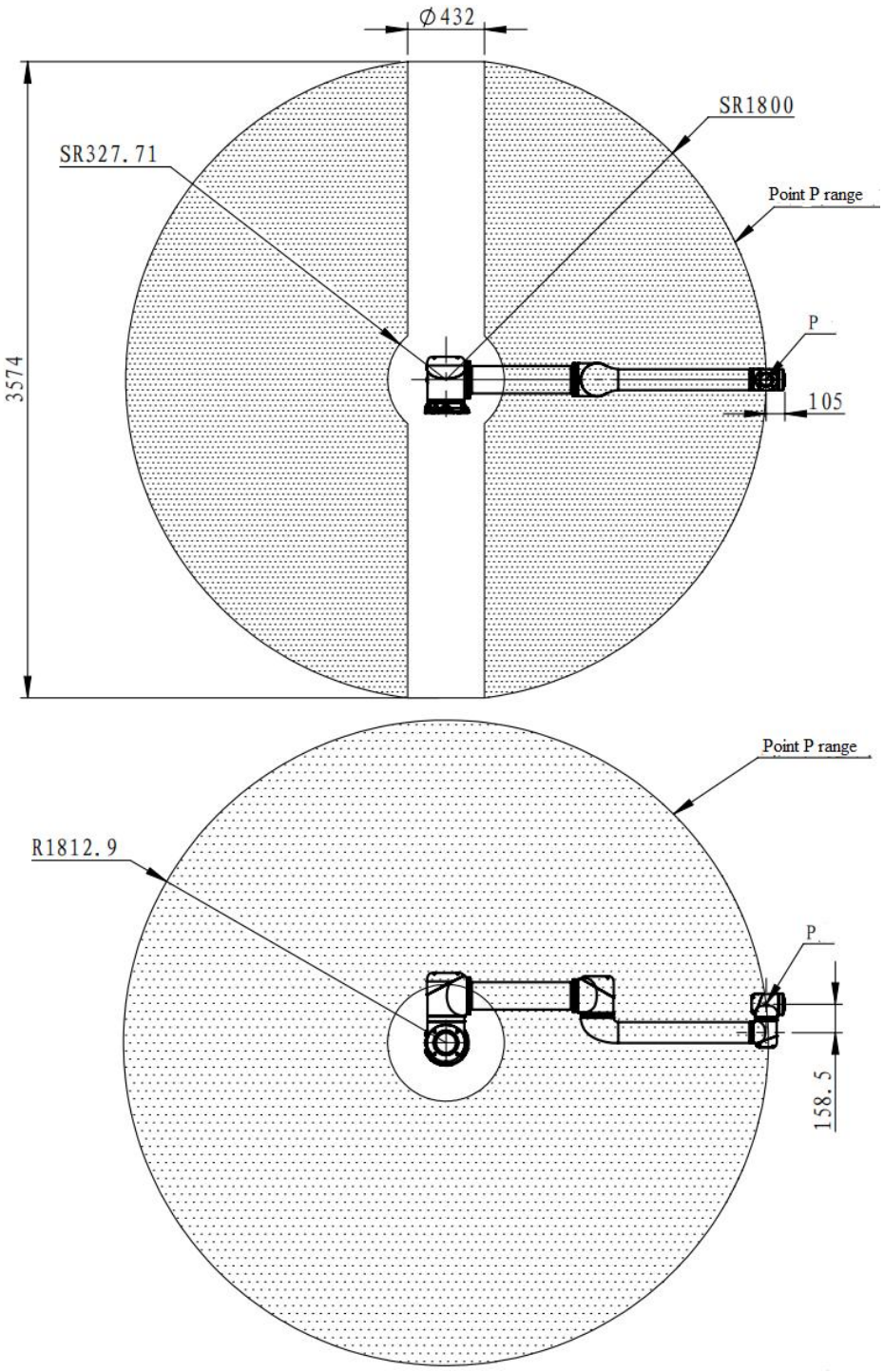
Movement Range of S20-1700 Robot



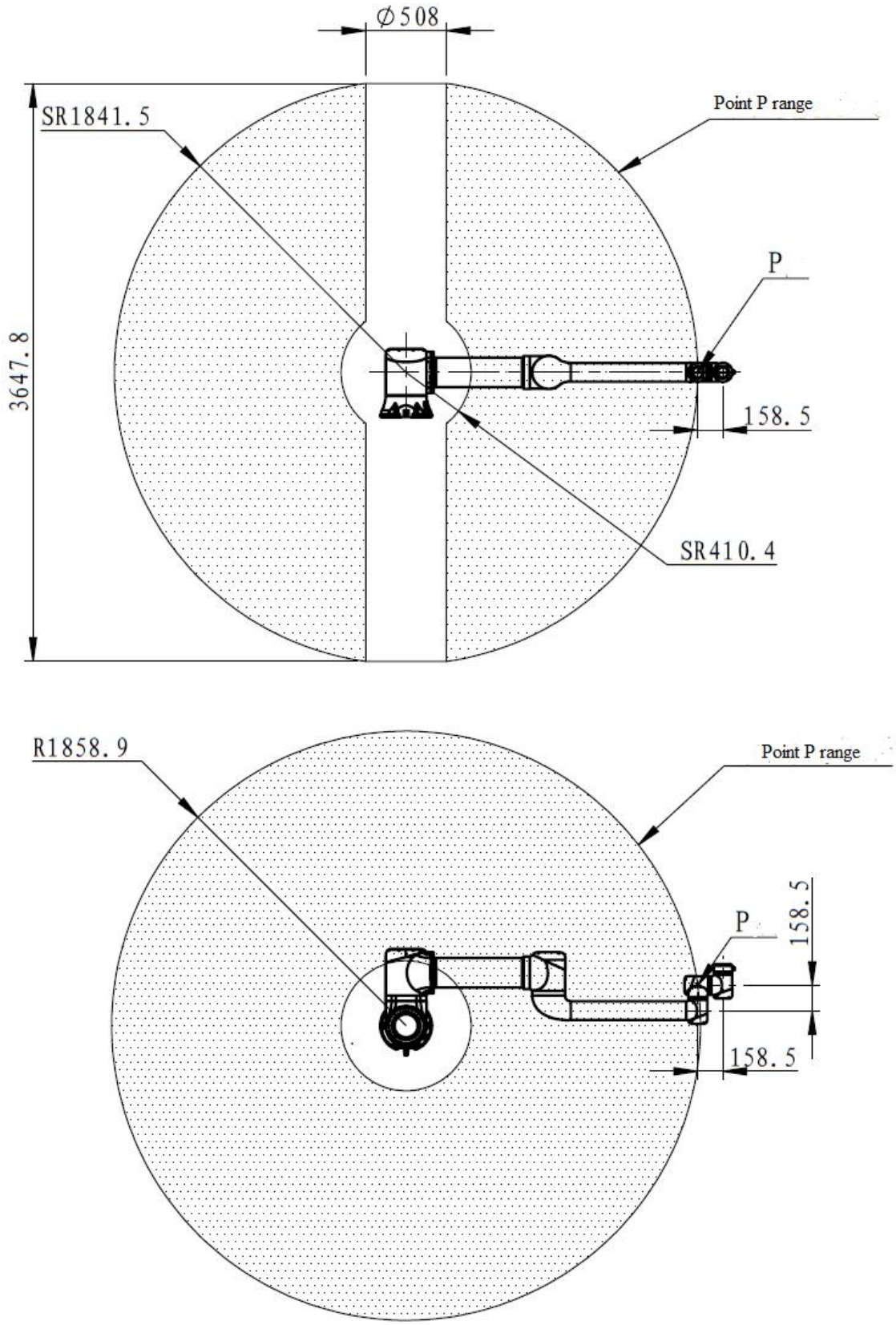
Movement Range of S30-1800 Robot



Movement Range of S35-1800 Robot



Movement Range of S40-1800 Robot



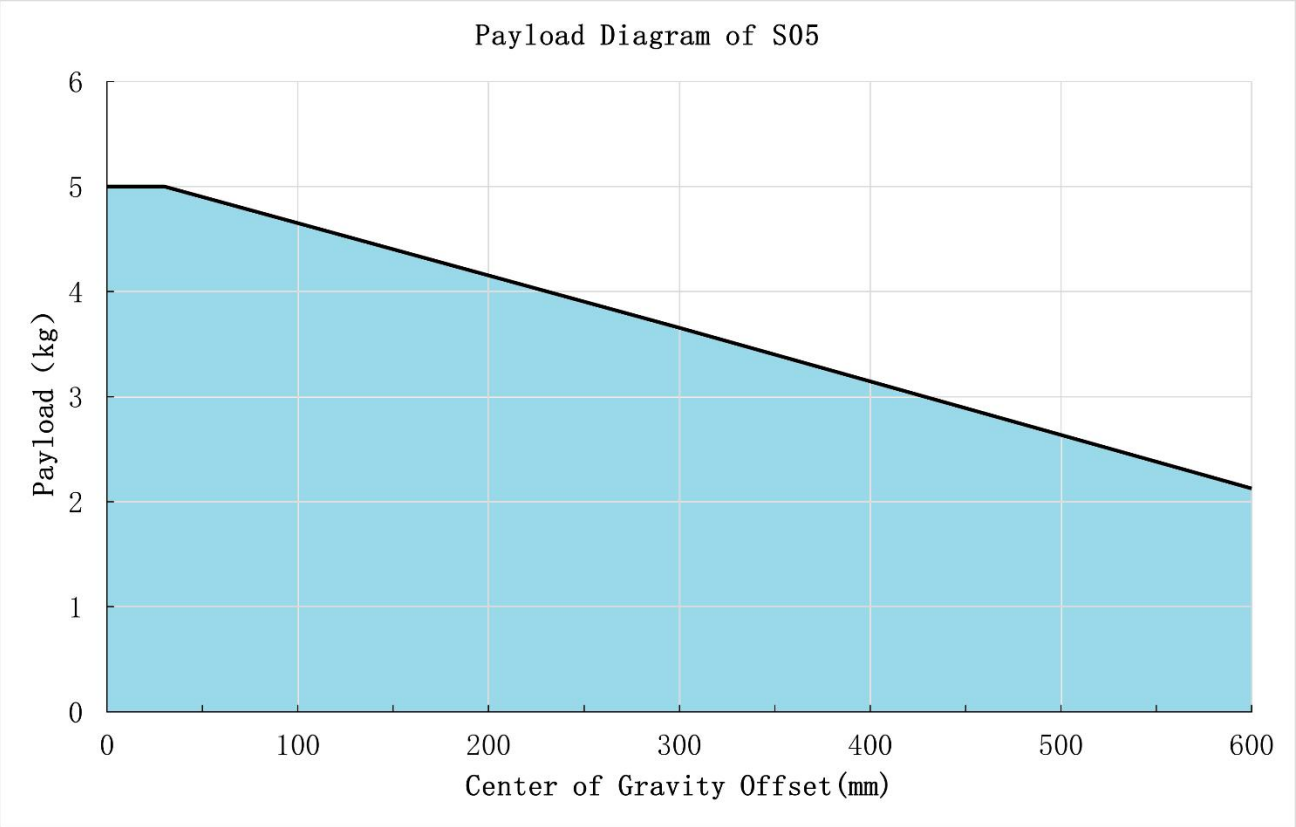
Movement Range of S50-2000 Robot

2.3.4 Maximum Robot Payload

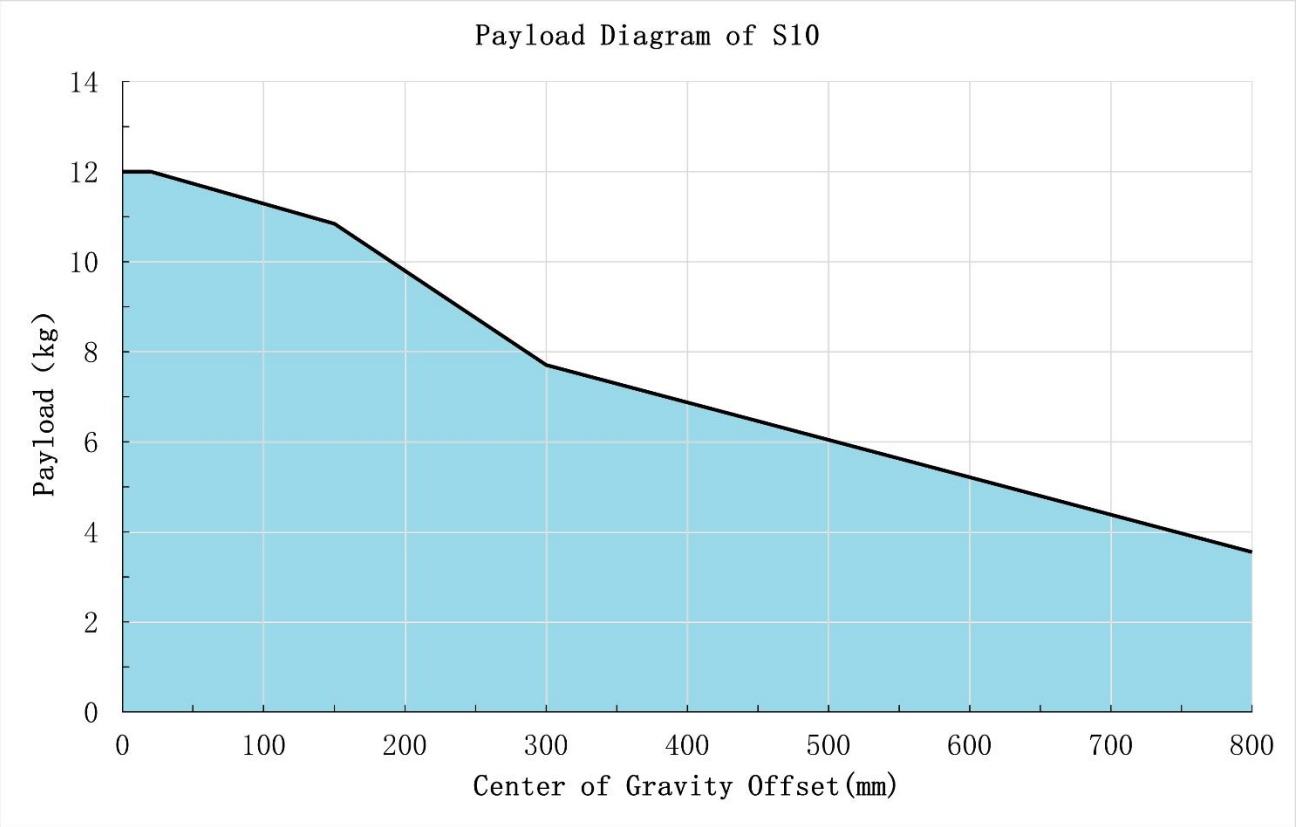
The maximum payload of the robot depends on the offset of the load's center of gravity (i.e. the distance between the load's center of gravity and the center of the tool flange, the load eccentricity). When calculating the payload in an application, the end tool and the objects that the tool grabs must be considered.

If the payload exceeds the maximum payload range, the ability of robot to accelerate may be reduced, or in more severe cases, the robot may become inoperable.

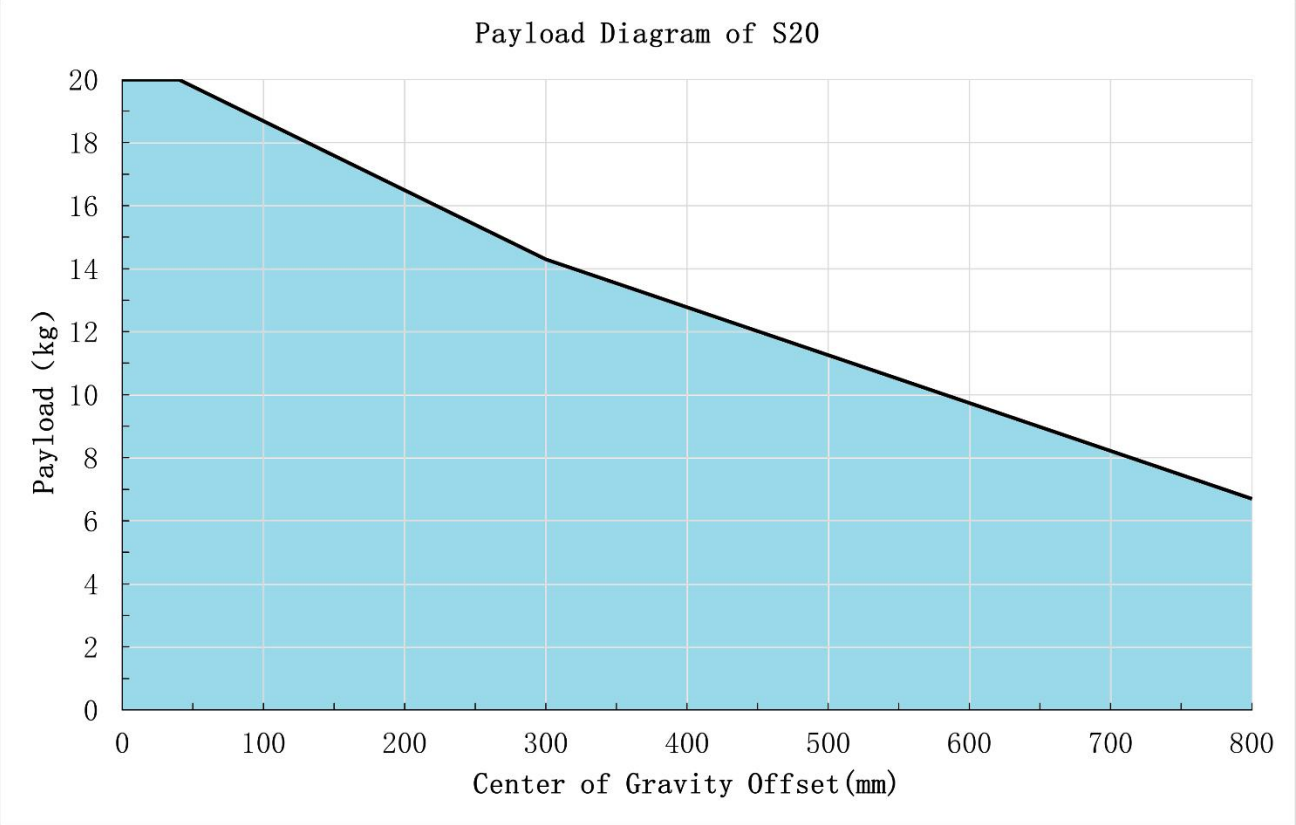
For payloads located under the tool flange (i.e., palletizing applications with the last joint facing down), the robot can accommodate larger payloads and load eccentricity. For example, for S20 and S30 robots, the load profile is plotted below:



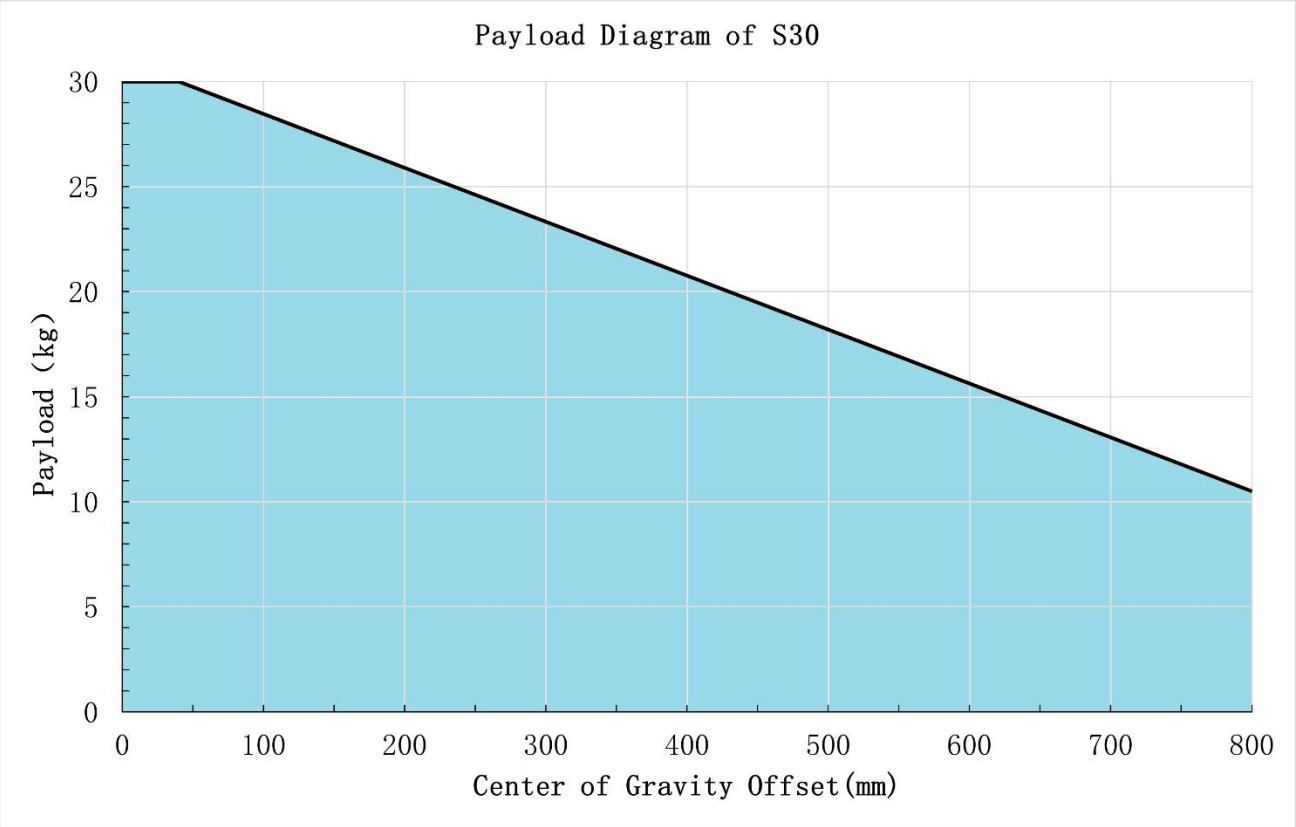
Payload Curve Diagram of S05-900 Robot



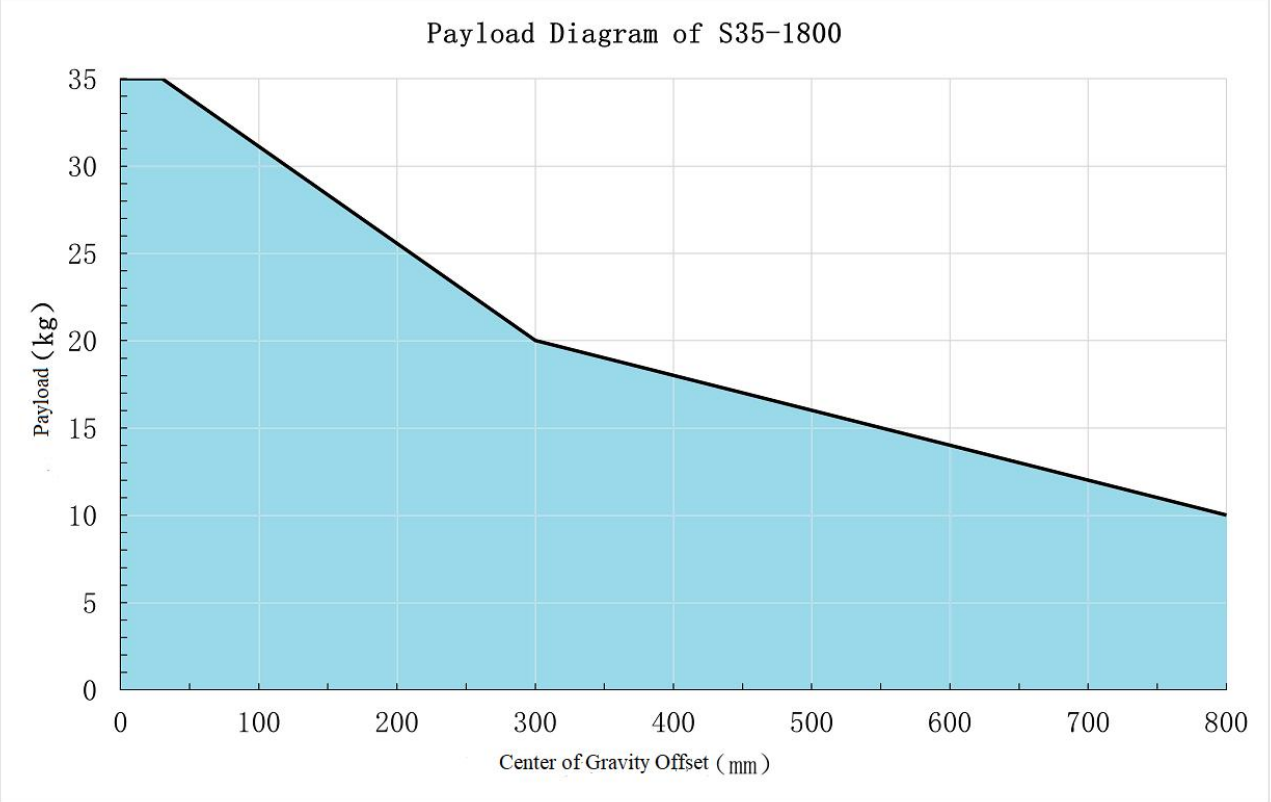
Payload Curve Diagram of S10-1300 Robot



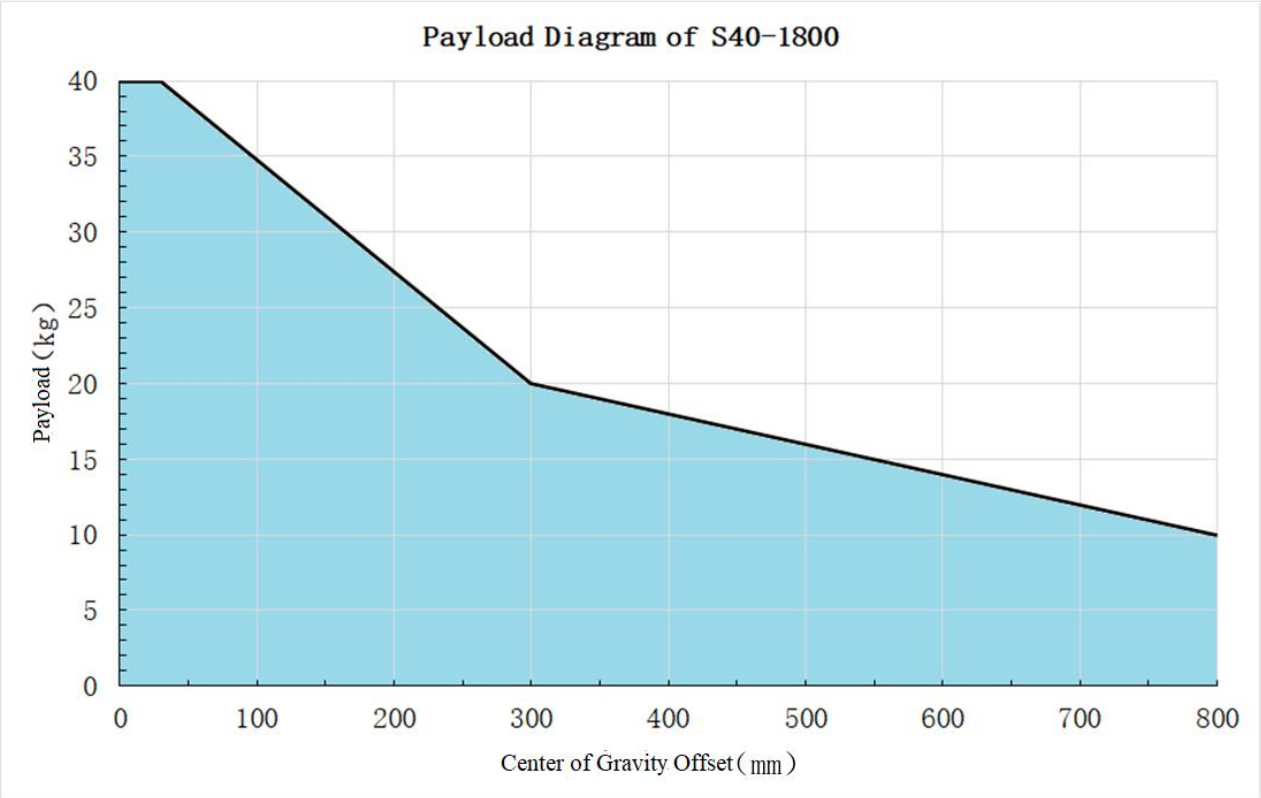
Payload Curve Diagram of S20-1700 Robot



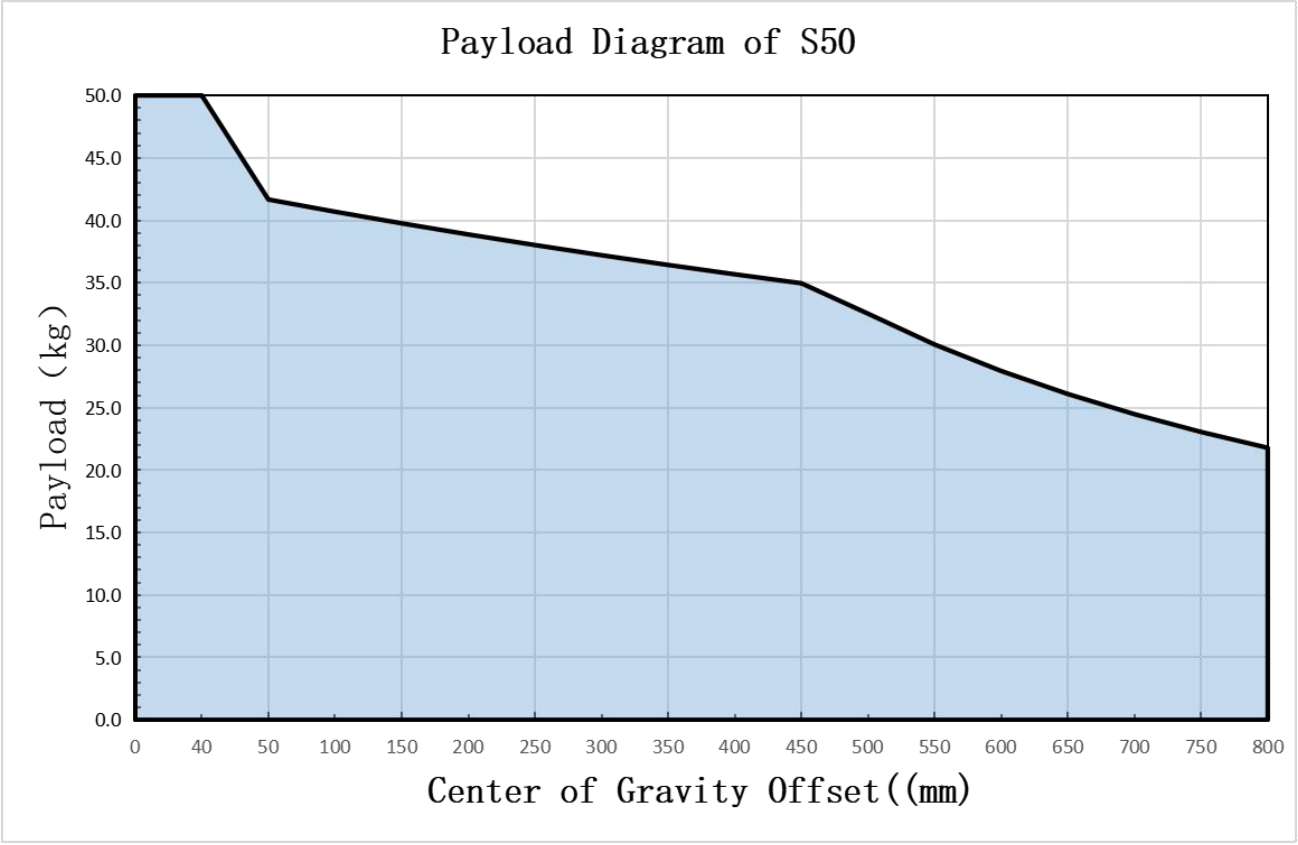
Payload Curve Diagram of S30-1800 Robot



Payload Curve Diagram of S35-1800 Robot



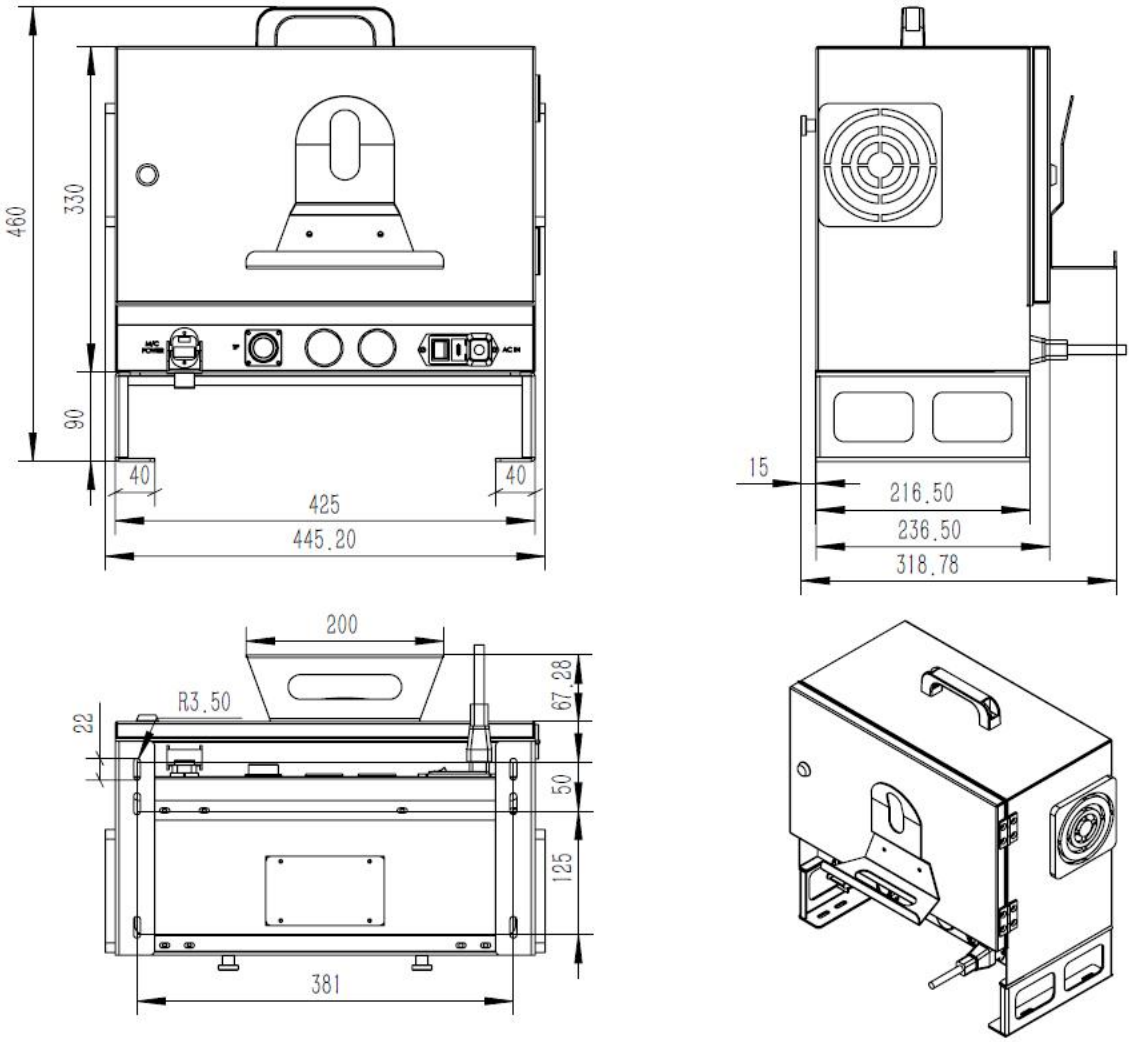
Payload Curve Diagram of S40-1800 Robot



Payload Curve Diagram of S50-2000 Robot

2.4 Robot Electrical Control Box

Dimension/Size

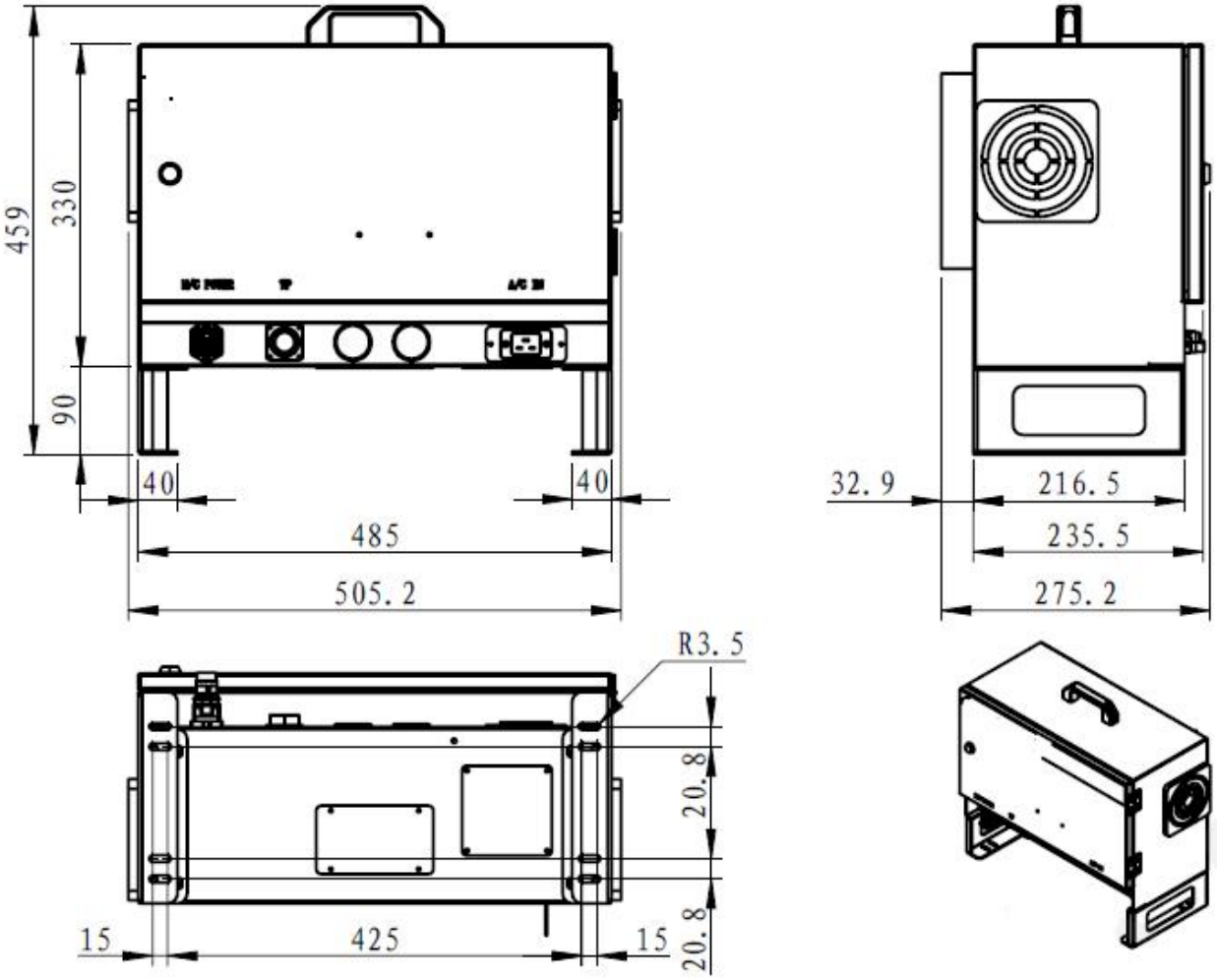


Technical Specification:

Technical Specifications:

Item	Specification
Weight	19.0 kg
Operating Temperature	0~50 ℃
Size of Electrical Control Box	536*445*319 mm
Paint Color	Dark Gray + White
External Power Input	100-240 V AC,50-60 Hz

Dimensions（S50）



Technical Specifications(S50):

Item	Specification
Weight	22.8Kg
Operating Temperature	0~50 ℃
Size of Electrical Control Box	505*459*275mm
Paint Color	Dark Gray + White
External Power Input	100-240V AC,50-60Hz

Chapter III Mechanical Installation

3.1 Unpacking

Step 1:

Open the carton and check that the contents match the items on the packing list. If any are missing or redundant, please contact your local sales office promptly.

Step 2:

Take out the robot and check that the robot is complete in appearance and that there are no obvious defects such as scratches. If any quality problems are found on appearance, please contact your local sales office promptly.

Step 3:

Carry the robot arm to the designated station for installation. The robot workspace is larger than its movement space (please refer to section 2.3.3 for robot movement space). When selecting a robot installation location, be sure to consider the space of the cylinder directly above and below the robot. It should be avoided as much as possible to move the tool close to the cylinder, otherwise, even a slow movement of the tool can lead to a fast movement of the robot joint, which can lead to crashes or other hazards.

Step 4:

For installation of the robot arm, please refer to the following section for the specifications of the screws to be used and the preload force.

Step 5:

Installation of the Control Box: The control box can be placed on the floor after it is removed from the packing box. A 50 mm gap should be left on each side of the control box to ensure that air can circulate freely and smoothly.

Step 6:

Installation of the Teach Pendent: The teach pendent can be hung on the wall or on the front panel bracket of the control box after it is removed to ensure that it will not fall.

Step 7:

Remove the robot arm cable and insert the aviation plugs at each end of the cable into the aviation plugs on the electrical control box and the robot arm. And test the connections for looseness and misplaced. After checking, plug the cable from the electrical control box of robot to the mains into the mains socket to complete the preparation for power-on and operation.

Warning:



- Avoid violent unpacking and take out the robot to prevent the unpacking process from causing scratches on the surface of the robot or impacts and collisions on the robot.
- Please unpack in a dry and clean external environment. When taking out the machine from the packing box, please note to protect the appearance of the robot from collision and scratch. Robots should be protected from collision during packaging and transport, which can lead to scratch on the surface or damage to the internal structure of the robot. It is recommended to use the original wooden case or foam of robot to wrap the robot surface during shipping.
- When lifting the robot, ensure that the lifting equipment can support the weight of the individual components of robot to prevent falls that could result in scratch on the robot surface, damage to its internal structure, and personnel injury or death. When handling the robot manually, the base of the robot should be dragged by hand and carried by two or more persons.
- When using other tools to carry the robot, ensure that the robot is secured to the carrying tool to prevent the robot from tipping over.

3.2 Installation Environment Requirements

The robot should be installed indoors and meet the following requirements:

- Room Temperature: 0-50 ℃ or 0-45 ℃, please refer to the technical data of the corresponding robot model (to prevent excessive temperature changes).
- Relative Humidity: 10-80% (no condensation).
- Avoid direct sunlight.
- Avoid dust, fume, salt, metal powder and water.
- Avoid shock and vibration.
- Keep away from flammable, explosive and corrosive gas-solid liquids.
- Keep away from sources of electrical interference.
- The range of installation space for the robot is detailed in the robot dimension drawing, and

the specification of electrical control box.

3.3 Installation Space

Please confirm the installation space with reference to the operating area of the robot and the dimensions of the electrical control box.

Warning:



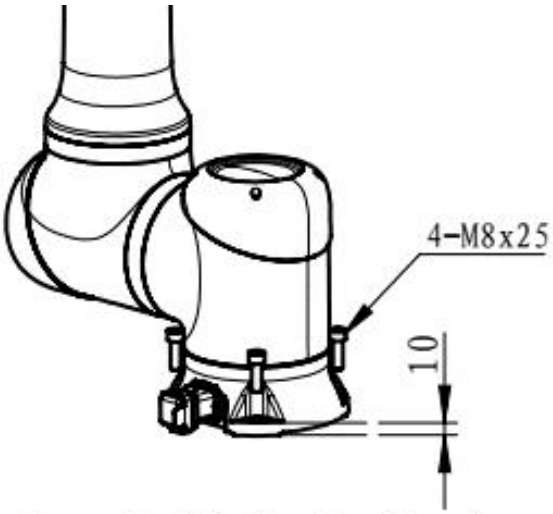
- In addition to the space required for installing robot and electrical control box, please ensure the minimum space required for the following conditions: Installation and workspace for robot end fixtures; Space for robot teaching; Activity space for installation, inspection and maintenance by the operator; Gas circuit space.
- The minimum bending radius of the robot power cable is 90 mm. When installing cables, be sure to leave enough installation space for power signal cables and other cables to prevent cables from being bent excessively.
- Ensure that the safety distance is at least 100 mm greater than the maximum operating space after the fixture is installed on the robot end-effector.

3.4 S05-900 Robot Arm Fixing

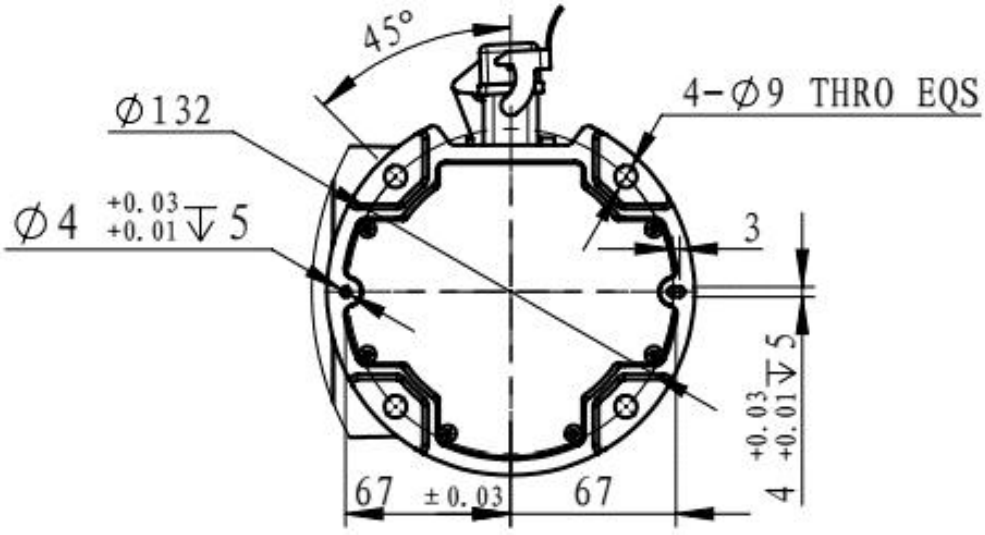
Secure the machine to the mounting abutment using four M8 (the specification is in accordance with performance class 10.9 or 12.9 labeled in ISO898-1) bolts and tighten the screws with a torque of 15Nm.

The robot is not supplied with a mounting abutment at the time of purchase. Please prepare or purchase a separate robot mounting abutment. The size and shape of the abutment varies depending on the robot system. However, these basic requirements should be observed:

- The thickness of the mounting abutment is at least 12mm, the steel plate is recommended to dampen the vibration.
- The roughness of the mounting surface of the mounting abutment is recommended to be 25μm and below.
- The base mounting dimensions of the robot are shown below and are measured in mm.



The threaded hole depth of mounting abutment is 12mm or more



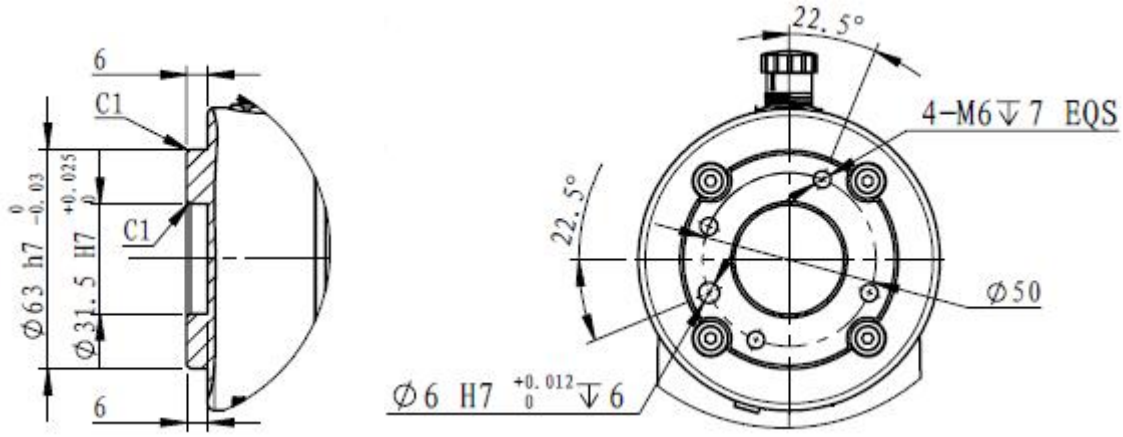
Mounting Dimension of S05-900 Robot Base

3.5 End Tool Mounting of S05-900 Robot

The robot is not supplied with end tools at the time of purchase, so please prepare or purchase them separately.

The threaded hole on the robot end for holding the tool is M6. When fixing the tool, please use the bolt with specification meeting the performance class 10.9 or 12.9 in ISO898-1, with a tightening torque of 15Nm.

The robot end mounting dimensions are shown below and are measured in mm.



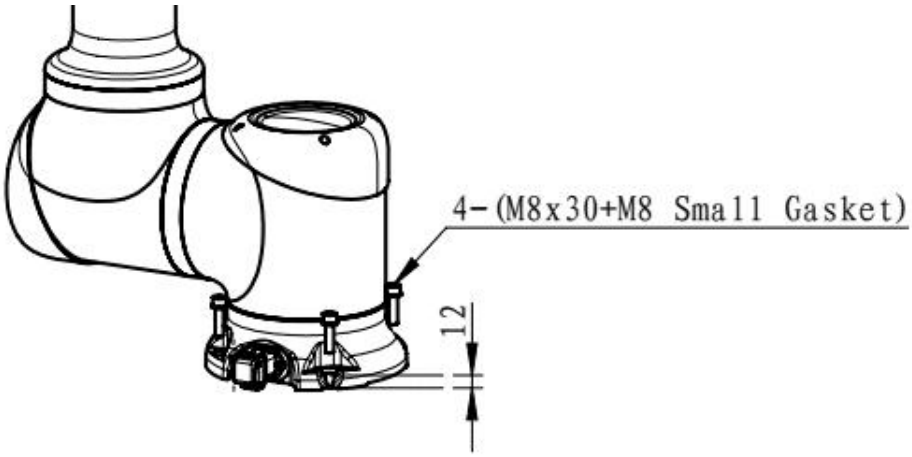
End flange dimension of S05-900 robot

3.6 S10-1300 Robot Arm Fixing

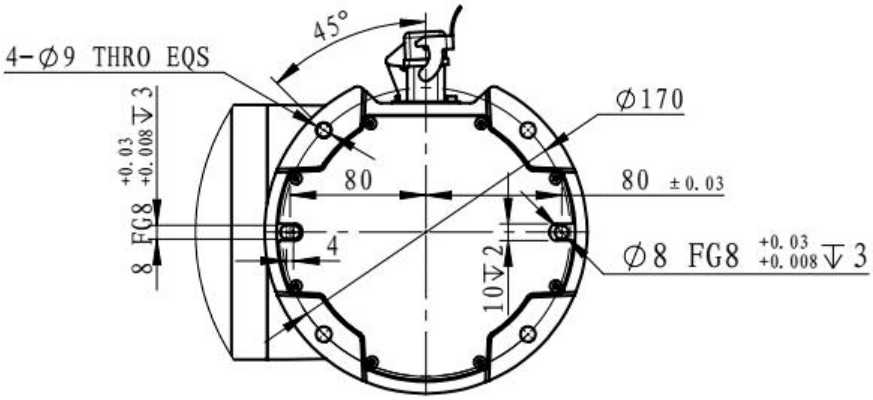
Secure the machine to the mounting abutment using four M8 (the specification is in accordance with performance classes 10.9 or 12.9 in ISO898-1) bolts and tighten the screws with a torque of 35Nm.

The robot is not supplied with a mounting abutment at the time of purchase. Please prepare or purchase a separate robot mounting abutment. The size and shape of the abutment varies depending on the robot system. However, these basic requirements should be observed:

- The thickness of the mounting abutment is at least 20 mm, the steel plate is recommended to dampen the vibration.
- The roughness of the mounting surface of the mounting abutment is recommended to be 25μm and below;
- Depth of threaded hole (M8) of mounting abutment is >16mm;



- The base mounting dimensions of the robot are shown below and are measured in mm;



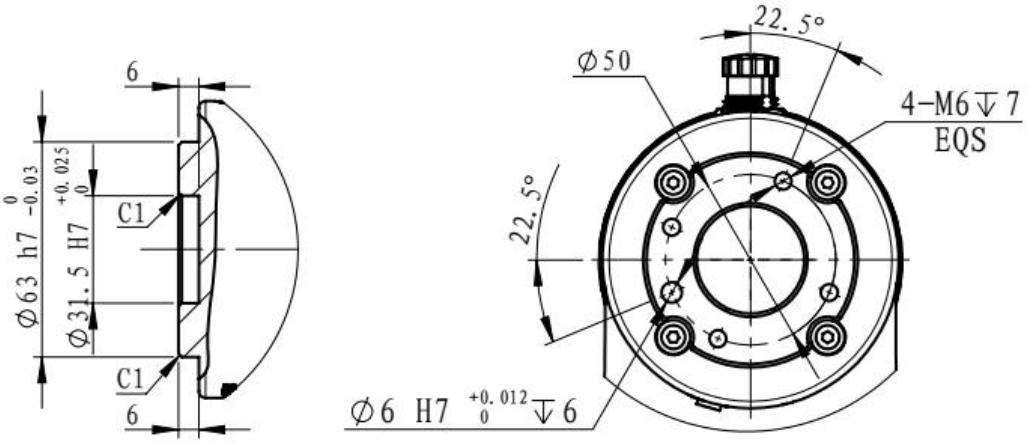
Mounting Dimension of S10-1300 Robot Base

3.7 End Tool Mounting of S10-1300 Robot

The robot is not supplied with end tools at the time of purchase, so please prepare or purchase them separately.

The threaded hole on the robot end for holding the tool is M6. When fixing the tool, please use the bolt with specification meeting the performance class 10.9 or 12.9 in ISO898-1, with a tightening torque of 15Nm.

The robot end mounting dimensions are shown below and are measured in mm.



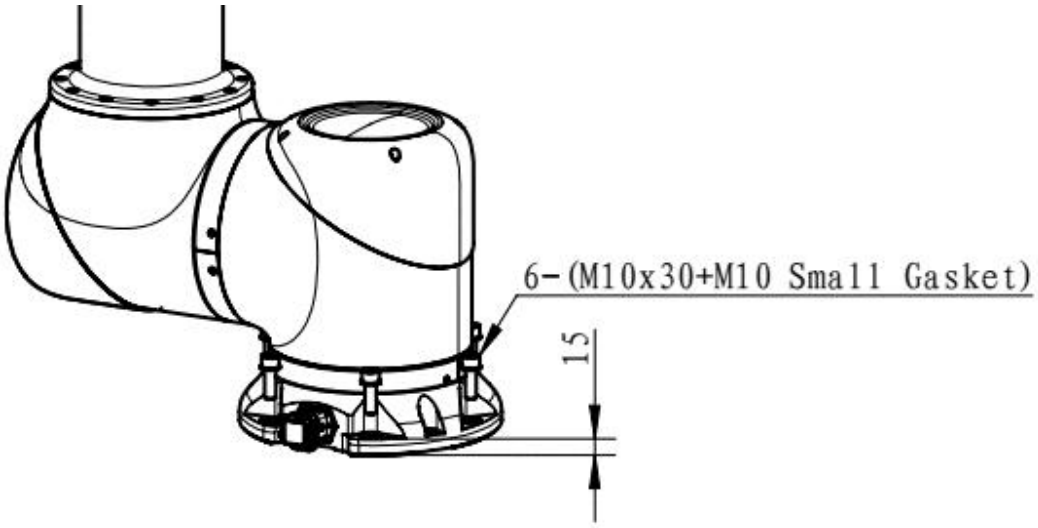
End Flange Dimension of S10-1300 Robot

3.8 S20-1700 Robot Arm Fixing

Please use the six M10 bolts (the specification is in accordance with performance classes 10.9 or 12.9 in ISO898-1) with M10 flat washer to secure the machine to the mounting abutment and use a torque of 74Nm to fasten the screws.

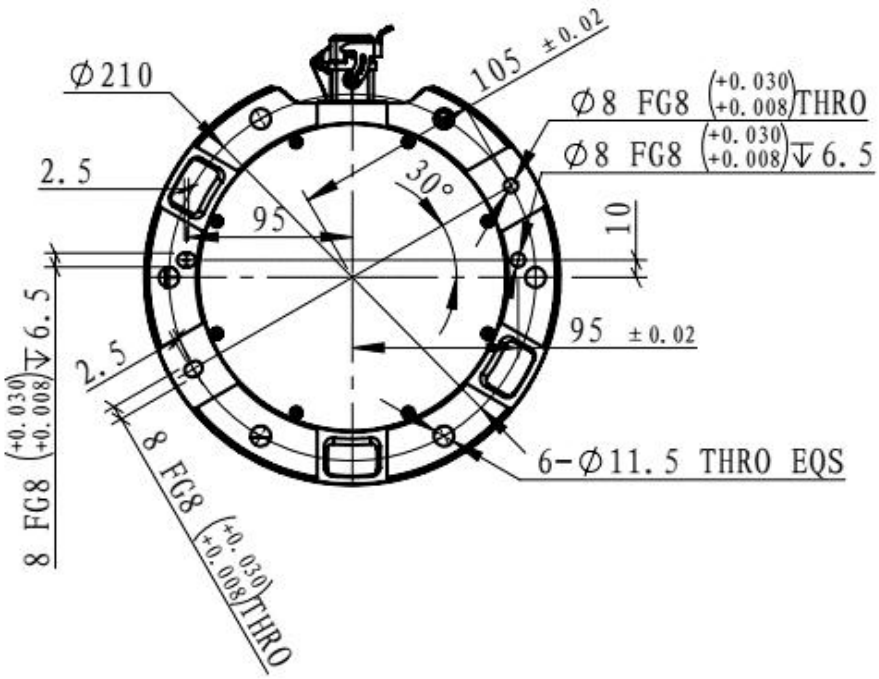
The robot is not supplied with a mounting abutment at the time of purchase. Please prepare or purchase a separate robot mounting abutment. The size and shape of the abutment varies depending on the robot system. However, these basic requirements should be observed:

- The thickness of the mounting abutment is at least 20 mm, the steel plate is recommended to dampen the vibration.
- The roughness of the mounting surface of the mounting abutment is recommended to be 25μm and below.



The depth of the threaded hole M10 of the mounting abutment is more than 20 mm

- The base mounting dimension of the robot is shown below and are measured in mm.



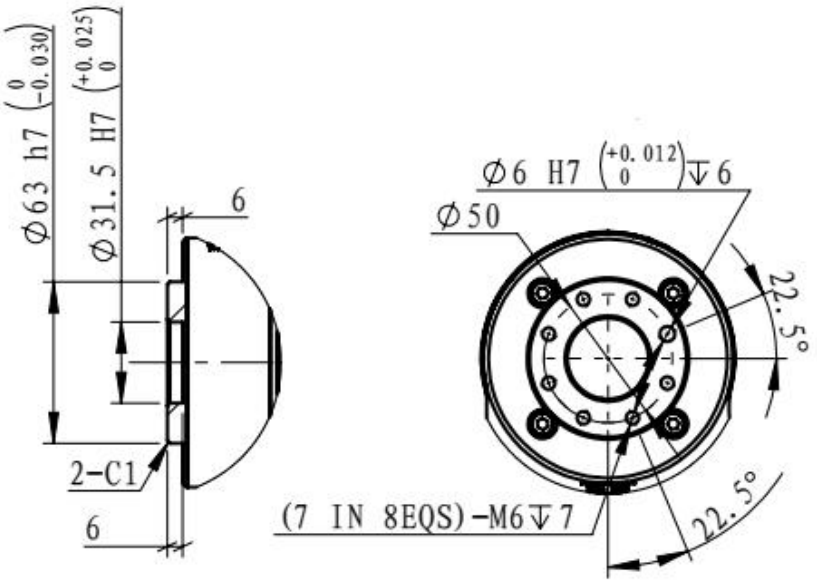
Mounting Dimension of S20-1700 Robot Base

3.9 End Tool Mounting of S20-1700 Robot

The robot is not supplied with end tools at the time of purchase, so please prepare or purchase them separately.

The threaded hole on the robot end for holding the tool is M6. When fixing the tool, please use the bolt with specification meeting the performance class 10.9 or 12.9 in ISO898-1, with a tightening torque of 15Nm.

The robot end mounting dimensions are shown below and are measured in mm.



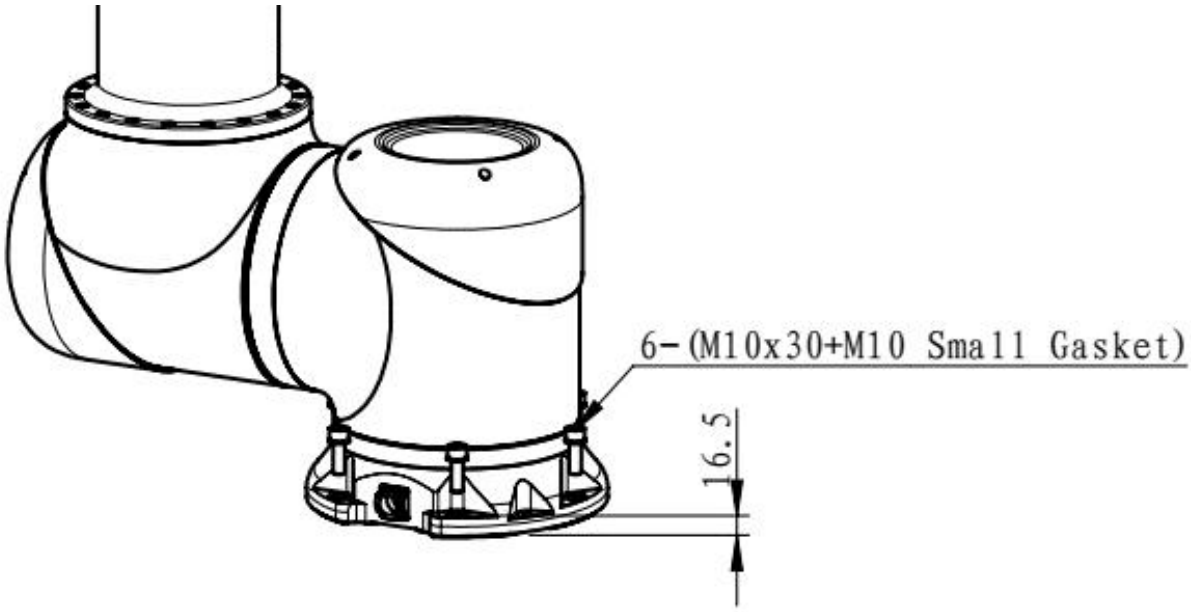
End Mounting Flange Dimensions of S20-1700 Robot

3.10 S30-1800 Robot Arm Fixing

Please use the six M10 bolts (the specification is in accordance with performance classes 10.9 or 12.9 in ISO898-1) with M10 flat washer to secure the machine to the mounting abutment and use a torque of 74Nm to fasten the screws.

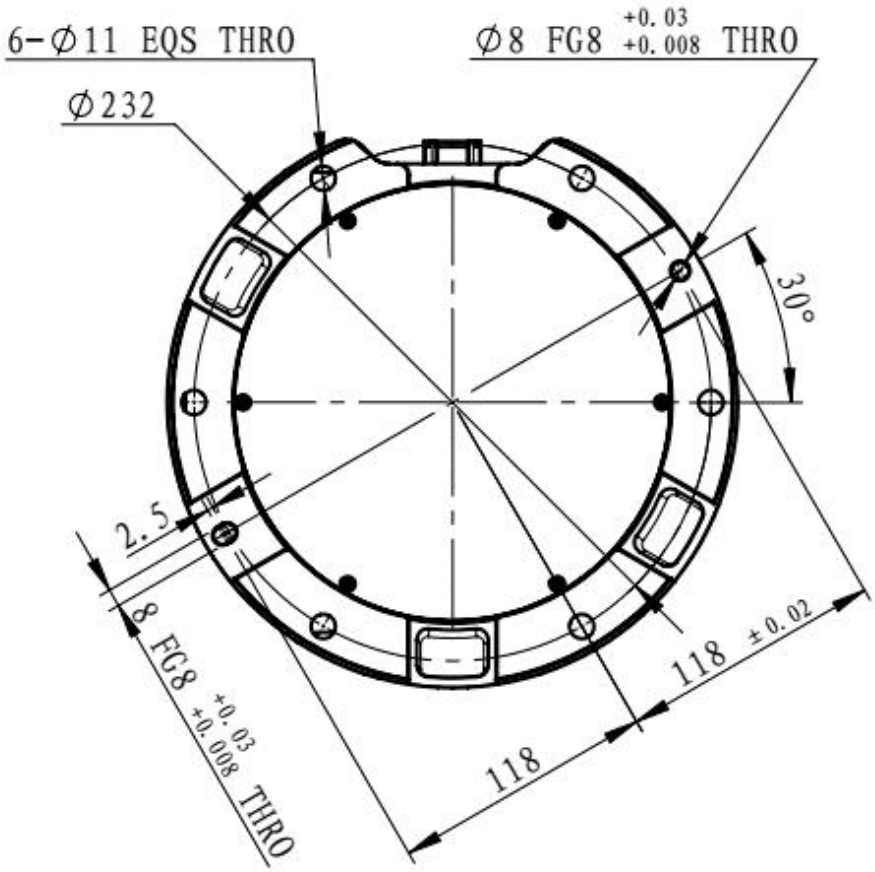
The robot is not supplied with a mounting abutment at the time of purchase. Please prepare or purchase a separate robot mounting abutment. The size and shape of the abutment varies depending on the robot system. However, these basic requirements should be observed:

- The thickness of the mounting abutment is at least 20 mm, the steel plate is recommended to dampen the vibration.
- The roughness of the mounting surface of the mounting abutment is recommended to be 25μm and below.



The depth of the threaded hole M10 of the mounting abutment is more than 20 mm

- The base mounting dimension of the robot is shown below and are measured in mm.



Mounting Dimension of S30-1800 Robot Base

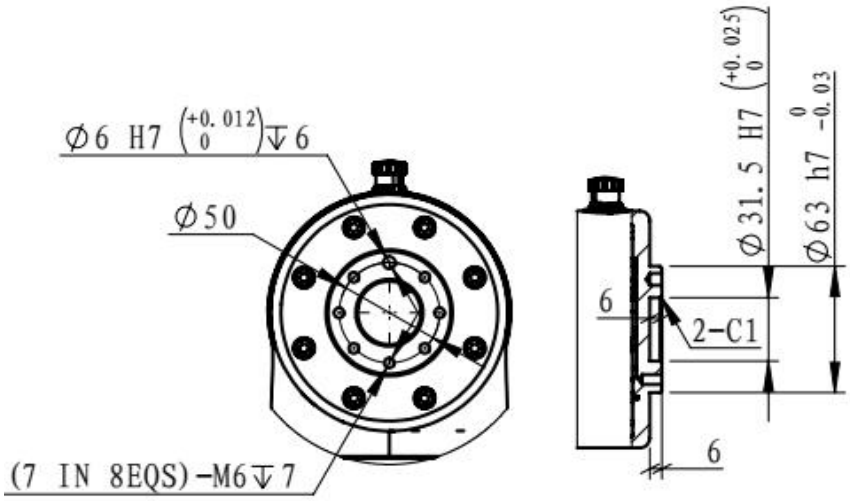
3.11 End Tool Mounting of S30-1800 Robot

The robot is not supplied with end tools at the time of purchase, so please prepare or purchase them

separately.

The threaded hole on the robot end for holding the tool is M6. When fixing the tool, please use the bolt with specification meeting the performance class 10.9 or 12.9 in ISO898-1, with a tightening torque of 15Nm.

The robot end mounting dimensions are shown below and are measured in mm.

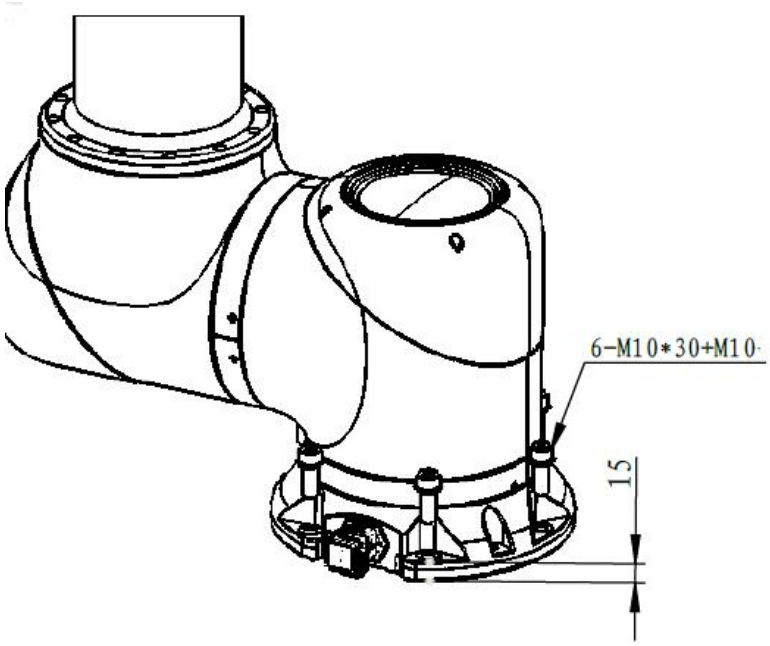


End Mounting Flange Dimension of S30-1800 Robot

3.12 S35-1800 Robot Arm Fixing

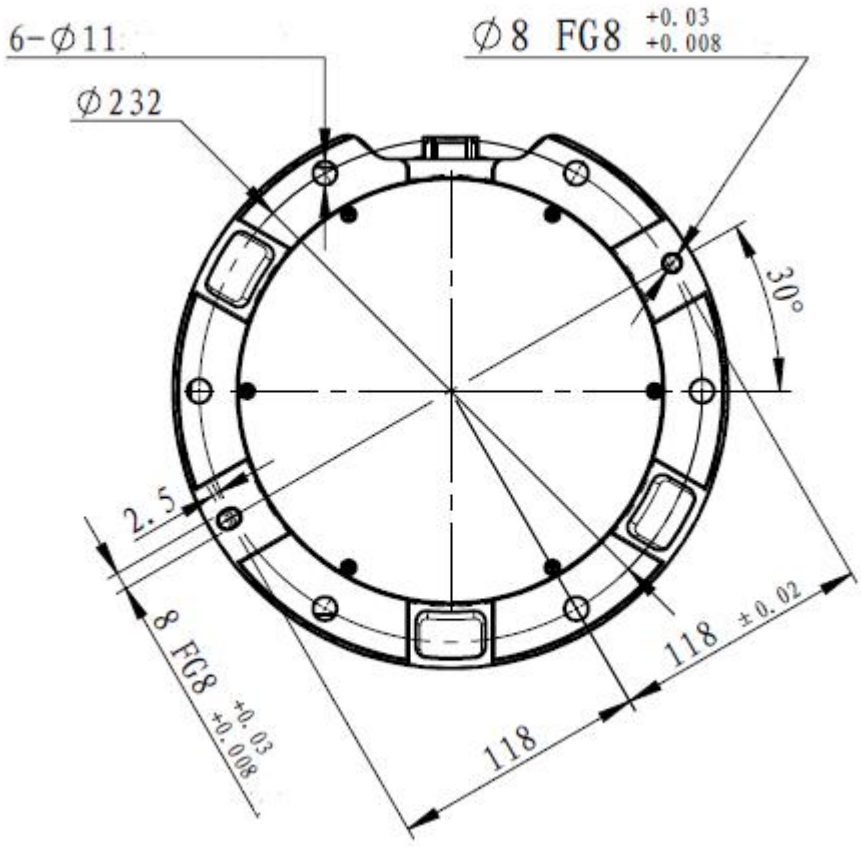
Please use the six M10 bolts (the specification is in accordance with performance classes 10.9 or 12.9 in ISO898-1) with M10 flat washer to secure the machine to the mounting abutment and use a torque of 74Nm to fasten the screws.

The robot is not supplied with a mounting abutment at the time of purchase. Please prepare or purchase a separate robot mounting abutment. The size and shape of the abutment varies depending on the robot system. However, these basic requirements should be observed:
The thickness of the mounting abutment is at least 20 mm, the steel plate is recommended to dampen the vibration.
The roughness of the mounting surface of the mounting abutment is recommended to be 25μm and below.



The depth of the threaded hole M10 of the mounting abutment is more than 20 mm

The base mounting dimension of the robot is shown below and are measured in mm.



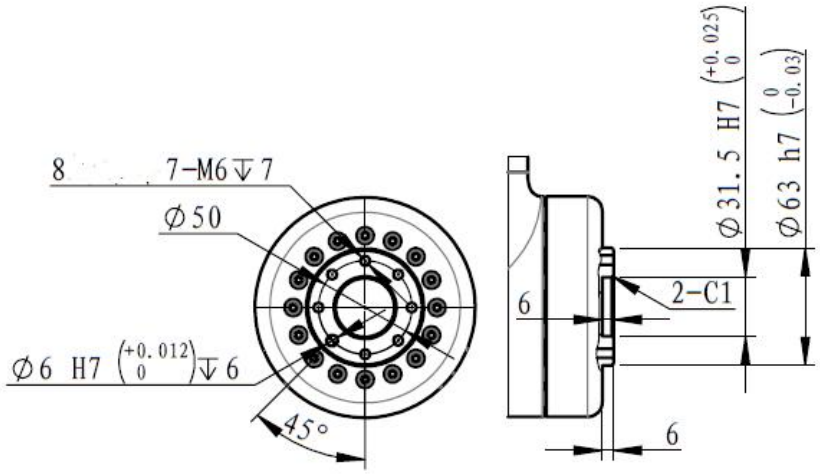
Mounting Dimension of S35-1800 Robot Base

3.13 End Tool Mounting of S35-1800 Robot

The robot is not supplied with end tools at the time of purchase, so please prepare or purchase them separately.

The threaded hole on the robot end for holding the tool is M6. When fixing the tool, please use the bolt with specification meeting the performance class 10.9 or 12.9 in ISO898-1, with a tightening torque of 15Nm.

The robot end mounting dimensions are shown below and are measured in mm

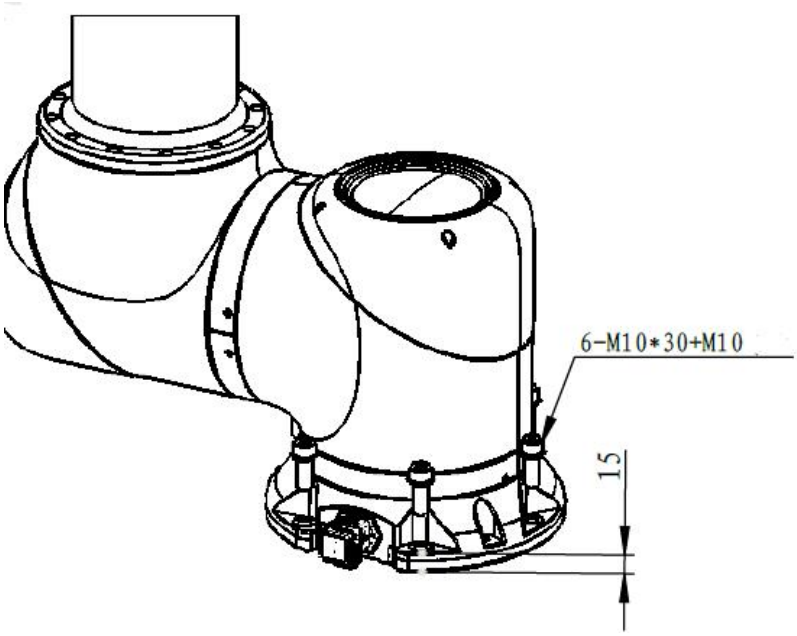


End Mounting Flange Dimension of S35-1800 Robot

3.14 S40-1800 Robot Arm Fixing

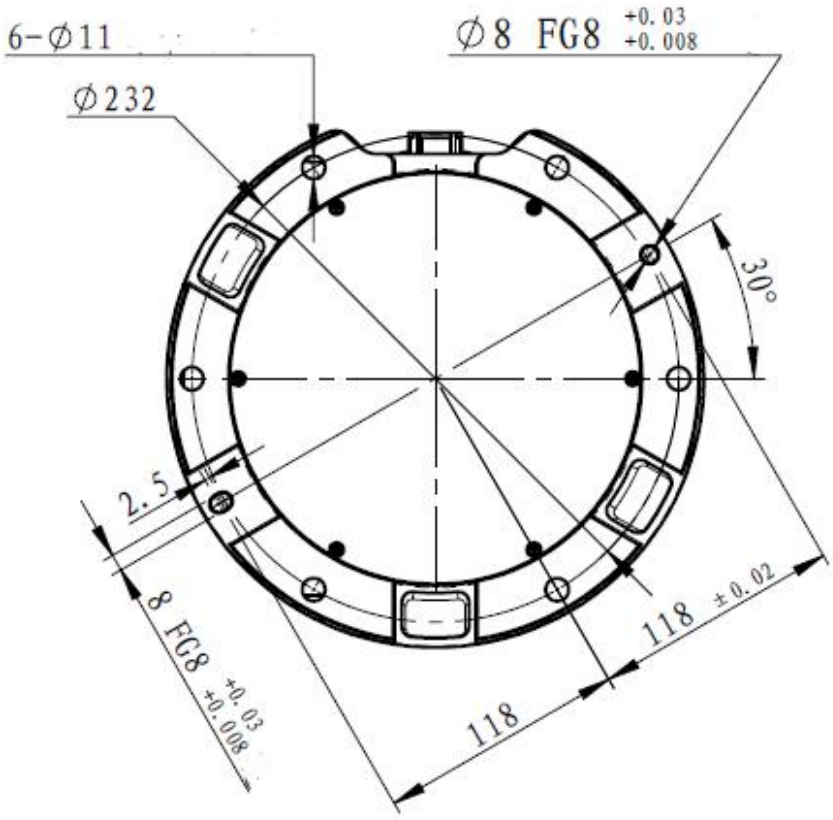
Please use the six M10 bolts (the specification is in accordance with performance classes 10.9 or 12.9 in ISO898-1) with M10 flat washer to secure the machine to the mounting abutment and use a torque of 74Nm to fasten the screws.

The robot is not supplied with a mounting abutment at the time of purchase. Please prepare or purchase a separate robot mounting abutment. The size and shape of the abutment varies depending on the robot system. However, these basic requirements should be observed:
The thickness of the mounting abutment is at least 20 mm, the steel plate is recommended to dampen the vibration.
The roughness of the mounting surface of the mounting abutment is recommended to be 25μm and below.



The depth of the threaded hole M10 of the mounting abutment is more than 20 mm

The base mounting dimension of the robot is shown below and are measured in mm.



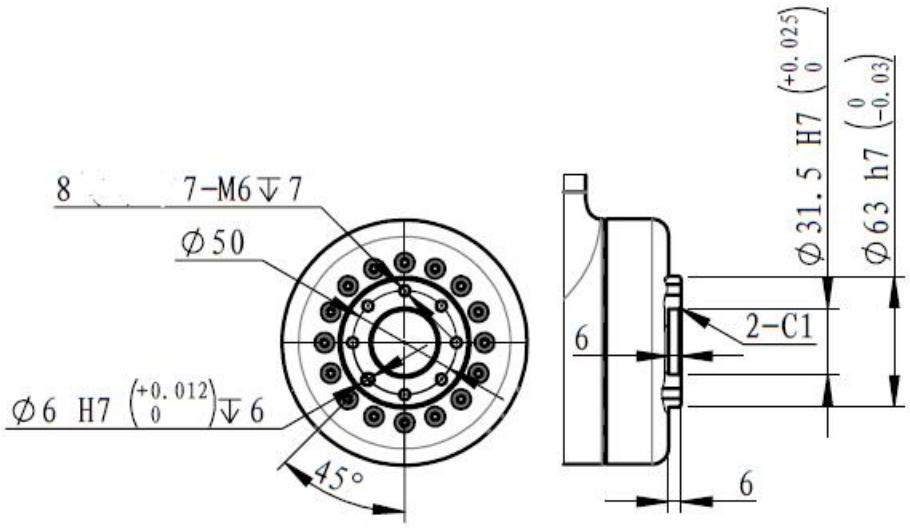
Mounting Dimension of S35-1800 Robot Base

3.16 End Tool Mounting of S40-1800 Robot

The robot is not supplied with end tools at the time of purchase, so please prepare or purchase them separately.

The threaded hole on the robot end for holding the tool is M6. When fixing the tool, please use the bolt with specification meeting the performance class 10.9 or 12.9 in ISO898-1, with a tightening torque of 15Nm.

The robot end mounting dimensions are shown below and are measured in mm。

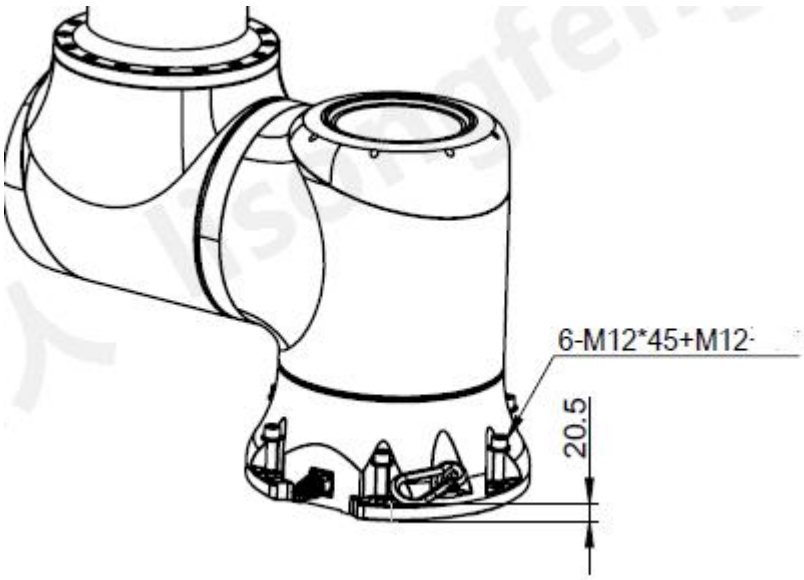


End Mounting Flange Dimension of S40-1800 Robot

3.16 S50-2000 Robot Arm Fixing

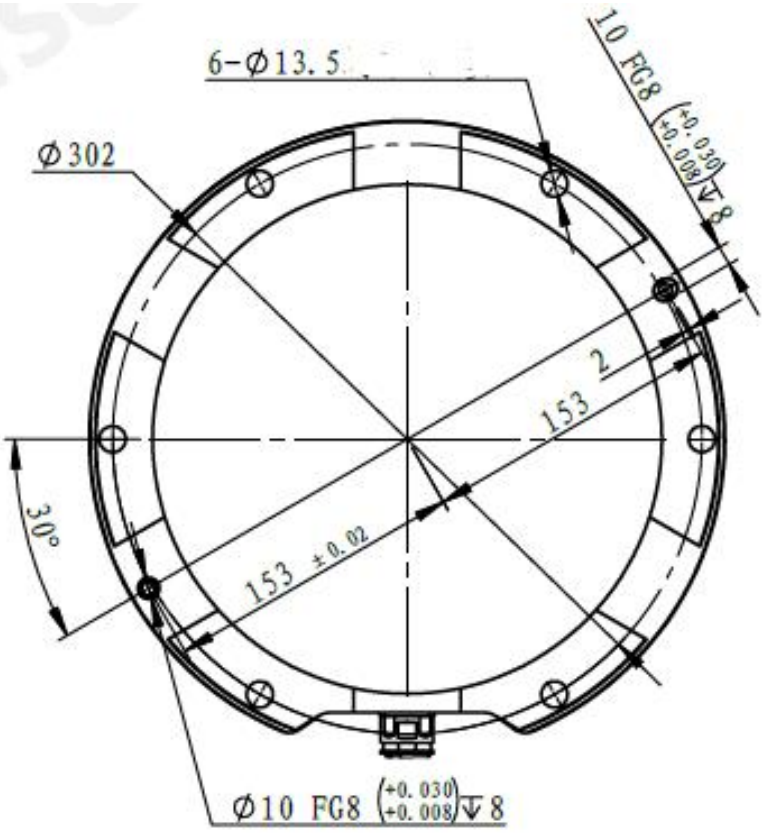
Please use the six M12 bolts (the specification is in accordance with performance classes 10.9 or 12.9 in ISO898-1) with M12 flat washer to secure the machine to the mounting abutment and use a torque of 133Nm to fasten the screws.

The robot is not supplied with a mounting abutment at the time of purchase. Please prepare or purchase a separate robot mounting abutment. The size and shape of the abutment varies depending on the robot system. However, these basic requirements should be observed:
The thickness of the mounting abutment is at least 25 mm, the steel plate is recommended to dampen the vibration.
The roughness of the mounting surface of the mounting abutment is recommended to be 25μm and below.



The depth of the threaded hole M12 of the mounting abutment is more than 25 mm

The base mounting dimension of the robot is shown below and are measured in mm。



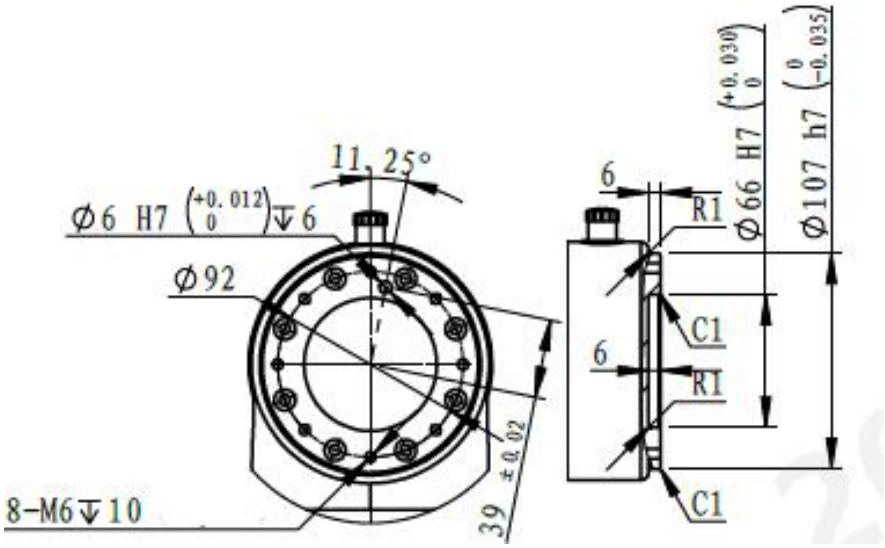
Mounting Dimension of S50-2000 Robot Base

3.17 End Tool Mounting of S50-2000 Robot

The robot is not supplied with end tools at the time of purchase, so please prepare or purchase them separately.

The threaded hole on the robot end for holding the tool is M6. When fixing the tool, please use the bolt with specification meeting the performance class 10.9 or 12.9 in ISO898-1, with a tightening torque of 15Nm.

The robot end mounting dimensions are shown below and are measured in mm。



End Mounting Flange Dimension of S50-2000 Robot

Chapter IV Quality Assurance

4.1 Product Quality Assurance

Without prejudice to any claim agreement that the user (customer) may have with a distributor or retailer, the manufacturer should give the customer a Product Quality Assurance in accordance with the terms set out below:

In the event of defects in the new equipment and its components due to faulty manufacture or materials within 12 months of commissioning (a maximum of 15 months if shipping time is included),Guangdong Huayan Robotics Co., Ltd.should provide the necessary spare parts and the user (customer) should provide the labor to replace the spare part, replace or repair the part in question with another part reflecting the latest state of the art. The Product Quality Assurance is null and void if the defective equipment is the result of improper handling or failure to follow the relevant information in the user manual. The Product Quality Assurance does not apply or extend to maintenance (e.g. installation, software downloads) performed by the authorized reseller or the customer. The user (customer) must provide the purchase receipt and date of purchase as valid evidence of the Product Quality Assurance. The claims under the Product Quality Assurance must be made within two months of the apparent failure to fulfil the Product Quality Assurance. The ownership of equipment or components that are replaced or returned toGuangdong Huayan Robotics Co., Ltd.is owned by Han's Robot. Any other claims arising out of or in connection with the equipment are not covered by the Product Quality Assurance. Nothing in the Product Quality Assurance attempts to limit or exclude the statutory rights of customer or the liability of manufacturer for injury or death caused by its negligence. The duration of the Product Quality Assurance should not be extended by the services provided under the terms of the Product Quality Assurance. Without prejudice to the principles of the Product Quality Assurance,Guangdong Huayan Robotics Co., Ltd.reserves the right to charge customers for replacement or repair costs. The above provisions are not intended to imply a change in the burden of proof to the detriment of the customer.

Han's Robot will not be liable for any damages or losses resulting from defective equipment, such as loss of production or damage to other production equipment.

4.2 Disclaimer

Han's Robot is committed to continually improving the reliability and performance of its products

and therefore reserves the right to upgrade its products without notice.Guangdong Huayan Robotics Co., Ltd.endeavors to ensure the accuracy and reliability of the contents of this manual, but cannot be held responsible for any errors or omissions in the information contained herein.

Chapter V Appendix

5.1 Technical Specification

Technical Specification of S05-900 Robot

Name	Parameter
Net Weight	23 kg
Payload	5 kg
Reach	900 mm
Joint Range	J1/J4/J5/J6: ±360° J2: -190°/+10° J3: ±158°
Joint Velocity	J1-J3 180 °/s J4-J6 200 °/s
Maximum Tool Speed	2 m/s
Repeatability Accuracy	±0.02mm
Degree of Freedom	6
Electrical Control Box Size	536*445*319 mm
End I/O Port	Digital Input 3
	Digital Output 3
	Analogue Input 2
Electrical Control Box I/O Port	Digital Input 16
	Digital Output 16
	Analogue Input 2
	Analogue Output 2
I/O Power Supply	24 V 2 A
Communication	TCP/IP and Modbus
Programming	Graphical Programming, Remote Procedure Call
IP Classification	IP54
Power Consumption	180 W Typical Application
Collaborative Operation	10 Advanced Safety Configuration Functions
Major Materials	Aluminum Alloy
Operating Temperature	0-50 °C
External Power Input	200~240 Vac, 50~60 Hz
Cable	Cable Connecting the Electrical Control Box: 5 m
	Cable Connecting the Teach Pendent: 5 m

Technical Specification of S10-1300 Robot

Name	Parameter
Net Weight	37 kg
Payload	12 kg
Reach	1300 mm
Joint Range	J1/J4/J5/J6: ±360° J2: -190°/+10° J3: ±167°
Joint Velocity	J1-J2 120 °/s J3 150 °/s J4-J6 180 °/s
Maximum Tool Speed	2 m/s
Repeatability Accuracy	±0.03mm
Degree of Freedom	6 Rotary Joints
Electrical Control Box Size	536*445*319 mm
End I/O Port	Digital Input 3
	Digital Output 3
	Analogue Input 2
Electrical Control Box I/O Port	Digital Input 16
	Digital Output 16
	Analogue Input 2
	Analogue Output 2
I/O Power Supply	24 V 2 A
Communication	TCP/IP and Modbus
Programming	Graphical Programming, Remote Procedure Call
IP Classification	IP54
Power Consumption	Typical Application 350 W
Collaborative Operation	10 Advanced Safety Configuration Functions
Major Materials	Aluminum Alloy
Operating Temperature	0-50 °C
External Power Input	200~240 Vac 50~60 Hz
Cable	Cable Connecting the Electrical Control Box: 5 m
	Cable Connecting the Teach Pendent: 5 m

Technical Specification of S20-1700 Robot

Name	Parameter
Net Weight	64 kg
Payload	20 kg
Reach	1700 mm
Joint Range	J1/J4/J5/J6: ±360° J2: -190°/+10° J3: ±168°
Joint Velocity	J1-J2 90 °/s J3 120 °/s J4-J6 180 °/s
Maximum Tool Speed	2 m/s
Repeatability Accuracy	±0.1 mm
Degree of Freedom	6 Rotary Joints
Electrical Control Box Size	536*445*319 mm
End I/O Port	Digital Input 3
	Digital Output 3
	Analogue Input 2
Electrical Control Box I/O Port	Digital Input 16
	Digital Output 16
	Analogue Input 2
	Analogue Output 2
I/O Power Supply	24 V 2 A
Communication	TCP/IP and Modbus
Programming	Graphical Programming, Remote Procedure Call
IP Classification	IP54
Power Consumption	Typical Application 800 W
Collaborative Operation	10 Advanced Safety Configuration Functions
Major Materials	Aluminum Alloy
Operating Temperature	0-45 ℃
External Power Input	200~240 Vac ,50~60 Hz
Cable	Cable Connecting the Electrical Control Box: 5 m
	Cable Connecting the Teach Pendent: 5 m

Technical Specification of S30-1800 Robot

Name	Parameter
Net Weight	93 kg
Payload	30 kg
Reach	1800 mm
Joint Range	J1/J4/J5/J6: ±360° J2: -190°/+10° J3: ±160°
Joint Velocity	J1-J2 120 °/s J3 120 °/s J4-J6 180 °/s
Maximum Tool Speed	2.5 m/s
Repeatability Accuracy	±0 .1 mm
Degree of Freedom	6 Rotary Joints
Electrical Control Box Size	536*445*319 mm
End I/O Port	Digital Input 3
	Digital Output 3
	Analogue Input 2
Electrical Control Box I/O Port	Digital Input 16
	Digital Output 16
	Analogue Input 2
	Analogue Output 2
I/O Power Supply	24 V 2 A
Communication	TCP/IP and Modbus
Programming	Graphical Programming, Remote Procedure Call
IP Classification	IP54
Power Consumption	Typical Application 1000 W
Major Materials	Aluminum Alloy
Operating Temperature	0-45 ℃
External Power Input	200~240 Vac ,50~60 Hz
Cable	Cable Connecting the Electrical Control Box: 5 m
	Cable Connecting the Teach Pendent: 5 m

Technical Specification of S35-1800 Robot

Name	Parameter
Net Weight	89 kg
Payload	35 kg
Reach	1800 mm
Joint Range	J1/J4/J5：±360° J2:-190°/+10° J3:±160°
Joint Velocity	J1-J3：120°/s J4-J5：180°/s
Maximum Tool Speed	3.7m/s
Repeatability Accuracy	±0 .05mm
Degree of Freedom	5 Rotary Joints
Electrical Control Box Size	536*445*319 mm
End I/O Port	/
	/
	/
Electrical Control Box I/O Port	Digital Input 16
	Digital Output 16
	Analogue Input 2
	Analogue Output 2
I/O Power Supply	24 V 2 A
Communication	TCP/IP and Modbus
Programming	Graphical Programming, Remote Procedure Call
IP Classification	IP54
Power Consumption	Typical Application 1000 W
Major Materials	Aluminum Alloy
Operating Temperature	0-45 ℃
External Power Input	200~240 Vac ,50~60 Hz
Cable	Cable Connecting the Electrical Control Box: 5 m
	Cable Connecting the Teach Pendent: 5 m

Technical Specification of S40-1800 Robot

Name	Parameter
Net Weight	89kg
Payload	40kg
Reach	1800mm
Joint Range	J1/J4/J5：±360° J2:-190°/+10° J3:±160°
Joint Velocity	J1-J3：120°/s J4-J5：180°/s
Maximum Tool Speed	3.7m/s
Repeatability Accuracy	±0 .05mm
Degree of Freedom	5 Rotary Joints
Electrical Control Box Size	536*445*319 mm
End I/O Port	/
	/
	/
Electrical Control Box I/O Port	Digital Input 16
	Digital Output 16
	Analogue Input 2
	Analogue Output 2
I/O Power Supply	24 V 2 A
Communication	TCP/IP and Modbus
Programming	Graphical Programming, Remote Procedure Call
IP Classification	IP54
Power Consumption	Typical Application 1000 W
Major Materials	Aluminum Alloy
Operating Temperature	0-45 ℃
External Power Input	200~240 Vac ,50~60 Hz
Cable	Cable Connecting the Electrical Control Box: 5 m
	Cable Connecting the Teach Pendent: 5 m

Technical Specification of S50-2000 Robot

Name	Parameter
Net Weight	156kg
Payload	50kg
Reach	2000mm
Joint Range	J1/J4/J5/J6：±360° J2:-190°/+10° J3:±160°
Joint Velocity	J1： 244°/s J2：150°/s J3：120°/s J4-J6：180°/s
Maximum Tool Speed	8.5m/s
Repeatability Accuracy	±0 .1mm
Degree of Freedom	6 Rotary Joints
Electrical Control Box Size	485*330*227mmm
End I/O Port	Digital Input 3
	Digital Output 3
	Analogue Input 2
Electrical Control Box I/O Port	Digital Input 16
	Digital Output 16
	Analogue Input 2
	Analogue Output 2
I/O Power Supply	24 V 2 A
Communication	TCP/IP and Modbus
Programming	Graphical Programming, Remote Procedure Call
IP Classification	IP54
Power Consumption	Typical Application 1000 W
Major Materials	Aluminum Alloy
Operating Temperature	0-45 ℃
External Power Input	200~240 Vac ,50~60 Hz
Cable	Cable Connecting the Electrical Control Box: 3 m
	Cable Connecting the Teach Pendent: 5 m

5.2 Restriction of Safety-related Functions

Restriction of Safety Functions	Description
Join Position	Minimum and maximum angular position of the joint
Joint Velocity	Maximum angular velocity of the joint
TCP Position	Plane in Cartesian space restricting the TCP position of the robot
TCP Velocity	Maximum TCP velocity of the robot
TCP Force	Maximum thrust of the robot TCP

5.3 Electrical Safety Specification

Robot Emergency Stop	Perform the type 1 shutdown* and notify the robot arm using the system emergency stop.
Emergency Stop Button	Perform the type 1 shutdown* and notify the robot arm using the system emergency stop.
System Emergency Stop	Perform the type 1 shutdown*
Protection Stop	Perform the type 2 shutdown*
Emergency Stop Reset	Recovery of robot arm from guard stop status

***Noun interpretation:**

- Type 0 stop: When the power supply of the robot arm is cut off, the robot arm stops operating immediately. This is an uncontrollable stop, and since each joint will brake as fast as it can, the robot arm may deviate from the programmed path.
  This protective stop can only be used when the safety evaluation limits have been exceeded or if there is an error in the safety evaluation part of the control system.
- Type 1 stop: When the robot arm is powered to stop it, the robot arm stops, and when the robot arm stops, the power is cut off. This is controlled stopping, where the robot arm follows a programmed path. The power is cut off after a second or once the robot arm is standing still.
- Type 2 stop: Controlled stop when the robot arm is powered on. The robot arm stops all movements in one second. Manipulation of the safety rating control system allows the robot arm to stay in the stopped position.